IMPERIAL
SCHOOL OF PUBLIC HEALTH

Infectious Disease Modelling

CID: 06031009 – *Addressing Type A Influenza Vaccination Prioritization in the UK*
04th January 2025

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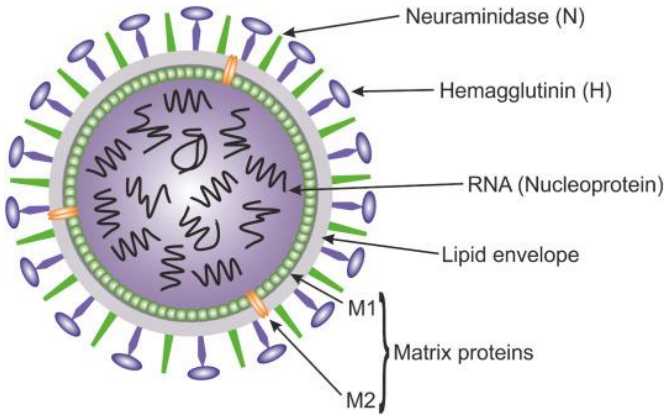
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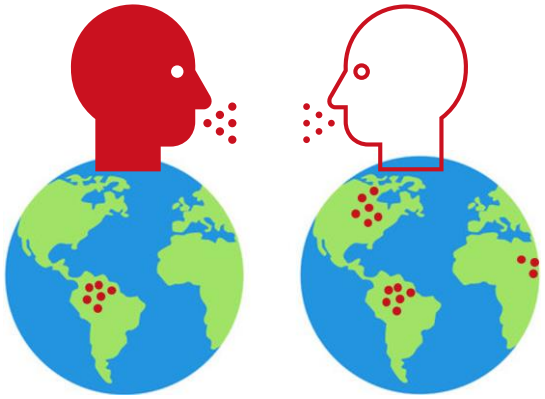
Influenza is an RNA Viral Respiratory Disease of Notable Public-Health Concern



Viral Transmission: Influenza in the Human Context

- **Seasonal influenza** is a costly disease both in terms of medical-cost and lost productivity
- Due to its ubiquity and contagiousness, it accounts for an **estimated annual global mortality of 250,000** to **500,000**, especially among persons with high-risk comorbidities / **high-risk groups (HRGs)**, **notably children under 5 and seniors** (1)
- Belonging to the family ***Orthomyxoviridae***, it is a non-cellular infectious virus consisting mainly of RNA enclosed by a lipoprotein envelope (2)
- There are **4 types** (A, B, C, and D), but **only A, B, and C infect humans** (2)
- Types **A and B are the only Influenza viruses known to cause disease pandemics or epidemics** (1)
- The most common type, A can be subclassified based on its two surface proteins: Hemagglutinin (**H**) which is responsible for host-cell interactions, and Neuraminidase (**N**) which are enzymes required for viral-replication (3)
- Type B can be classed based on its H surface proteins (3)

Influenza's epidemiology calls for vaccination to prevent pandemics & epidemics



Epidemic Pandemic

Viral Transmission: Seasonality & Vaccination

- These viruses can be **transmitted up to 6ft away primarily via droplets** made when an infected person coughs, sneezes, or talks (2)
- **Influenza A(H1N1), A(H3N2), and B-viruses co-circulate globally**, with outbreaks typically being sporadic and seasonal during early spring, late fall, and winter seasons (4)
- **New A and B variants emerge due to genetic mutations and recombination events** that occur during viral-replication, resulting in 'antigenic' shifts and drifts that serve as the basis for its seasonal nature
- These epidemiological considerations underline the importance of routine and universal vaccination in defence against outbreaks through the creation of **herd immunity**
- However, **budget-constraints often limit the ability of people in positions of power to actualize such broad immunization programs**
- For this reason, powerful stakeholders must determine who to prioritize for vaccination to best benefit the public

Influenza Type A(H1N1)'s infection in humans can be divided into three blocks

Infection 1

- For this purpose, we possess an understanding of influenza's life cycle, which can be used to provide a theoretical basis for public health decision-making
- Once inside the body, influenza **targets cells in the upper respiratory tract** and binds to and enters host cells using their H proteins (2,3)

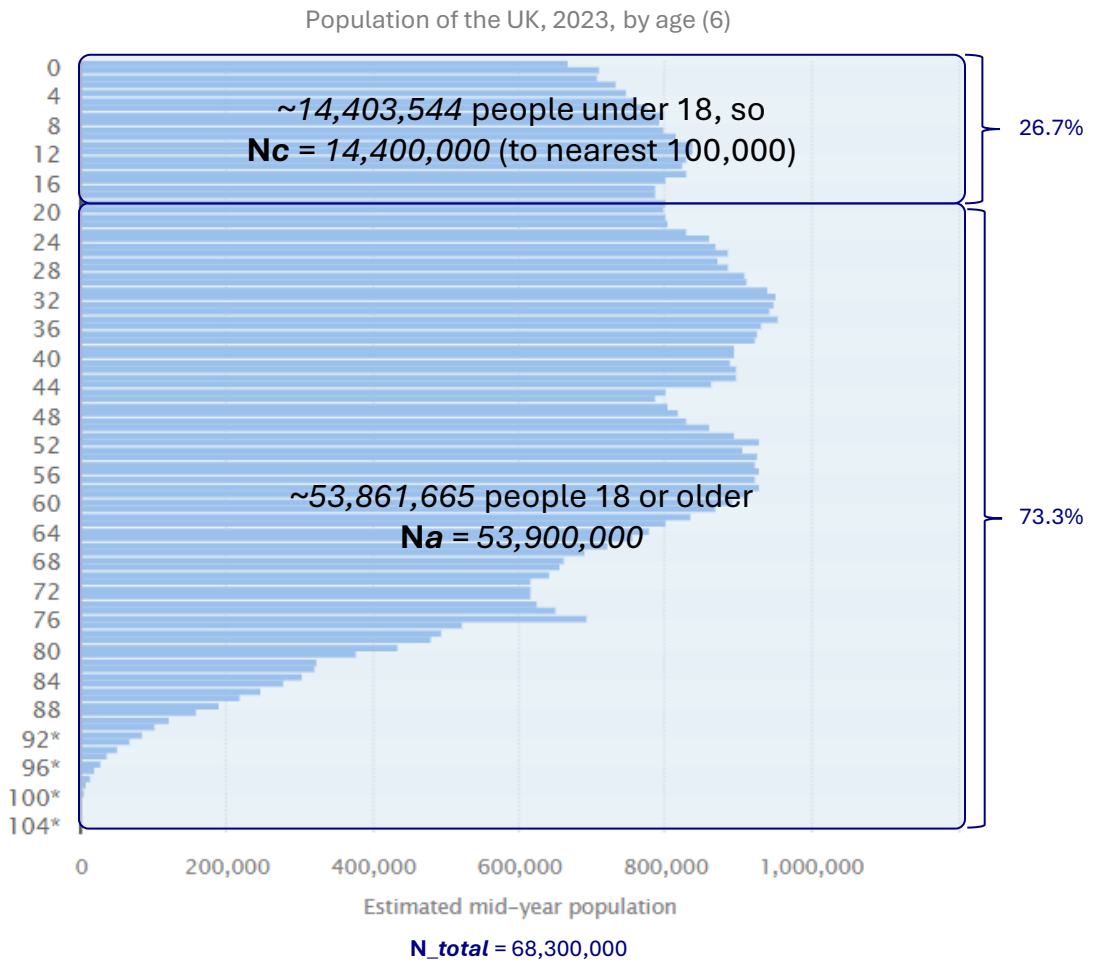
Latency & Infectious Periods 2

- They replicate using their N proteins, eventually bursting host-cells to release new viruses (Appendix A)
- Symptoms and inflammation begins once the immune system detects the infection (5)
- For A(H1N1), one of the most notable Influenza strains, the **mean latency period is 1.6 days** (5)
- This period refers to the time from infection until a person becomes infectious

Death or Recovery 3

- If the infection is successful, the host will die
- However, **most people recover in 5 days** as the immune system clears the virus

The UK's research & experience with Influenza, & demographics, call for disease modelling



Age in the spread and outcome of infection

Studies have indicated that children are disproportionately responsible most of influenza transmission, while the influenza mortality burden falls disproportionately on the elderly

Diversity & Demographics

Diverse populations including various age groups, health conditions, and socioeconomic backgrounds necessitate analysis that spans wider determinants of health

Historical Experience

The UK has faced notable pandemics (i.e. H1N1 2009); robust historical data can inform decisions on whether to prioritize children or the elderly based on risks of disease and transmission dynamics

Public Health Infrastructure

Its **well-established health system** facilitates effective data collection that can be used to inform prioritization preceding a pandemic

Influenza modelling can be used to simulate future outbreaks based on vaccine prioritizations

Objectives

1. Build a model to explore to demonstrate the importance of population vaccination
2. Use the same model to explore a scenario where children are prioritized for vaccination
3. Use the same model to explore a scenario where adults are prioritized for vaccination
4. Discuss herd immunity thresholds for both scenarios to determine which group to prioritize
5. Explore scenarios where antiviral treatment is introduced

Research Question: *"Considering the UK's population subject to a single wave of pandemic A(H1N1) Influenza, what group (children under 18, or adults over 18) should we prioritize for vaccination preceding pandemic, given we only have enough vaccine to cover 20% of the population?"*

For the sake of simplicity, we can take data averages; actual figures will include SDs & vary year to year

The A(H1N1) Vaccine

- There is only enough vaccine **to cover 20% of the total population**, which is 13,660,000 out of 68,300,000 people
- This is equivalent to **95% of the under 18 population**, or **25% of the 18+** population

- The vaccine provides **60% protection** against infection to all vaccinees in the context of a UK population

Mean Number of Contacts

- Every 1 child on average contacts 10 other children, and 3 adults per day
- Every 1 adult on average contacts 5 other adults, and 2 children per day

Probability of Infection

- Studies indicate the **risk of infection by Type A** in household contacts between 38% and 50% when one member is infected and infectious (7,8)
- While this may differ from risk of infection in day-to-day life, we will take the mean, **44%**, to address our question

Mean Latency Period

- Influenza A(H1N1)'s mean latency period is **1.6 days**

Recovery Rates

- All individuals (child and adult) who where infected and has not died will recover within an average of **5 days**

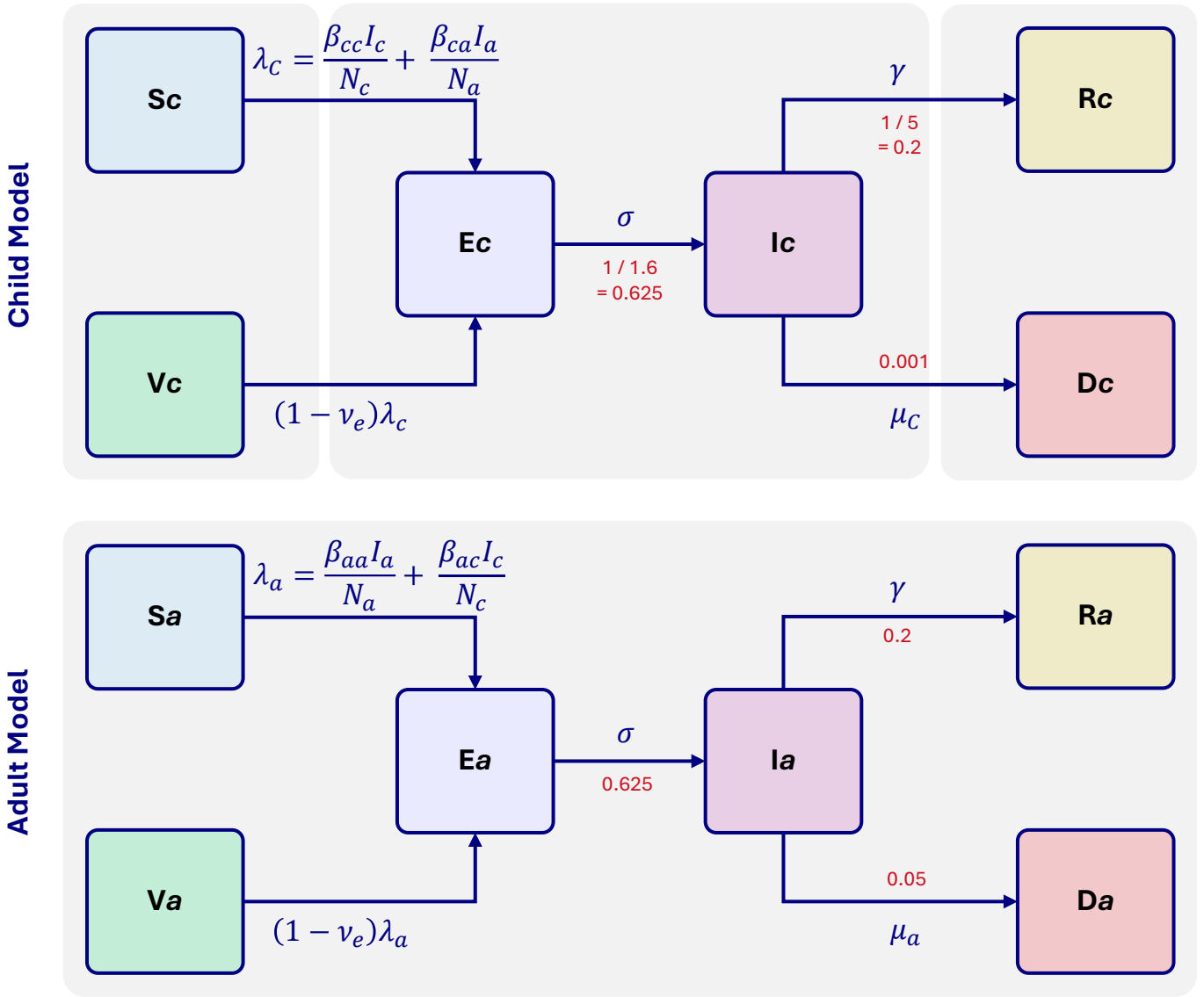
Death Rates

- **0.1% of infected children will die** from disease
- **5% of infected adults will die** from disease

Infection Rate

β = mean no. of contacts × assumed probability of infection

Influenza's epidemiology calls for vaccination to prevent pandemics & epidemics



Rate Definitions & Parameters

x_c = Children

x_a = Adults

Beta *cc* = The infection rate between children (4.4)

Beta *aa* = The infection rate between adults (2.2)

Beta *ca* = The infection rate from adults to children (0.88)

Beta *ac* = The infection rate from children to adults (1.32)

Lambda = The force of infection (FOI)

Nue = The vaccine's effectiveness (0.6)

Sigma = The progression rate from exposed to infectious

Gamma = The recovery rate

Mu = Death rates

S = Susceptible

I = Symptomatic, infected, and infectious

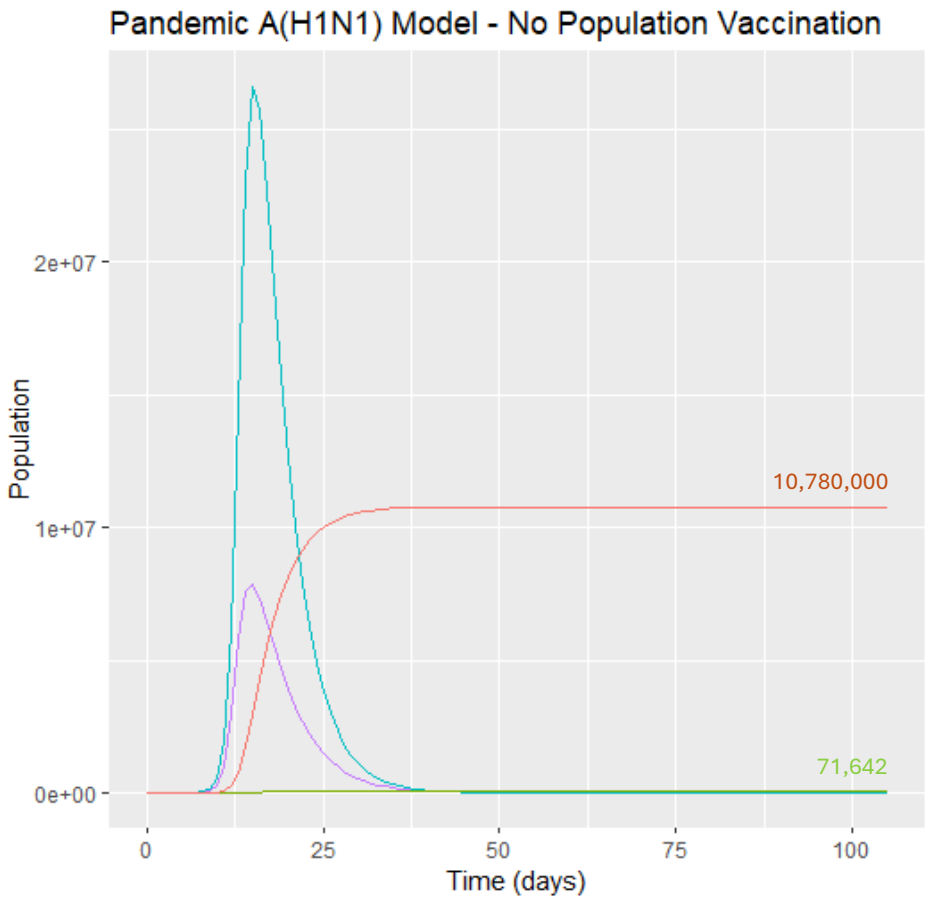
E = Latent; Exposed but not infectious

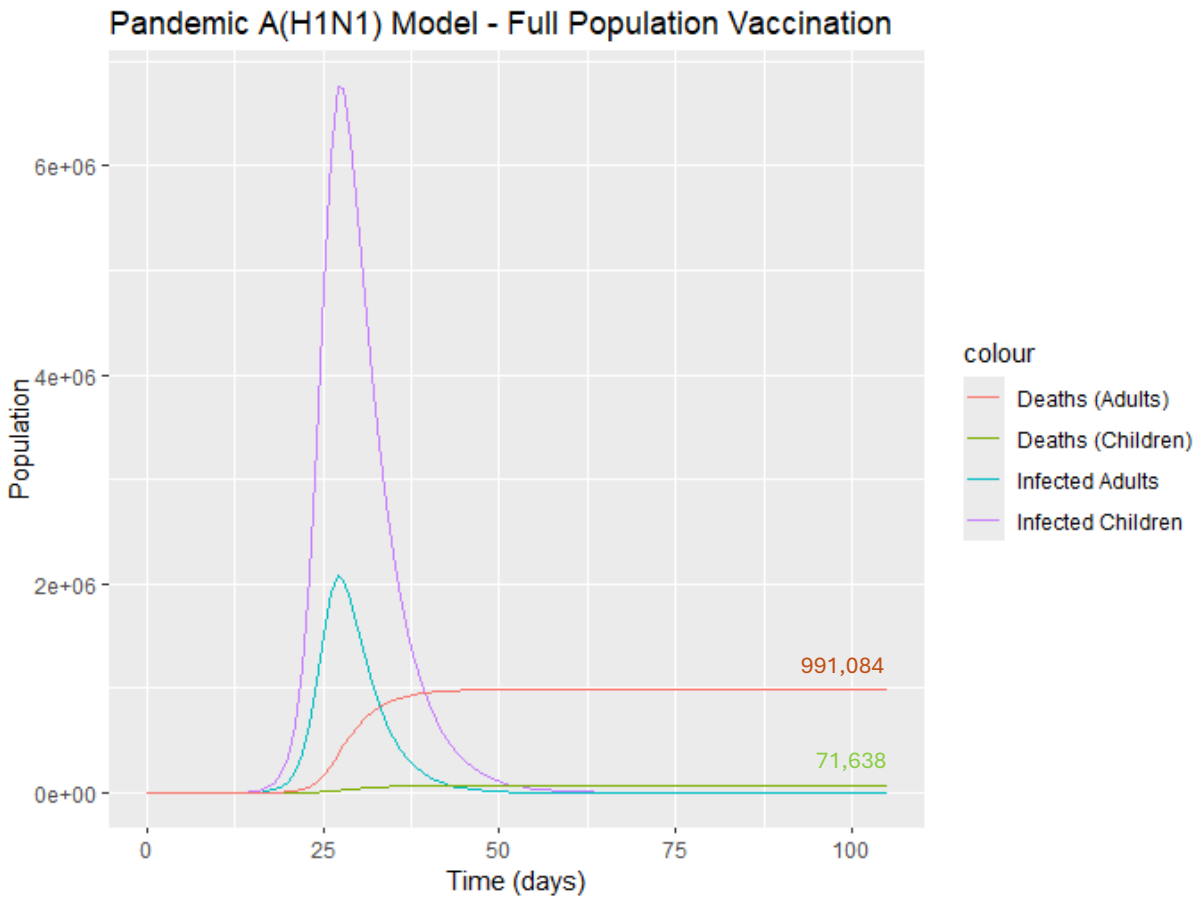
D = Deceased due to infection

V = Vaccinated; partially protected

R = Recovered; fully immune

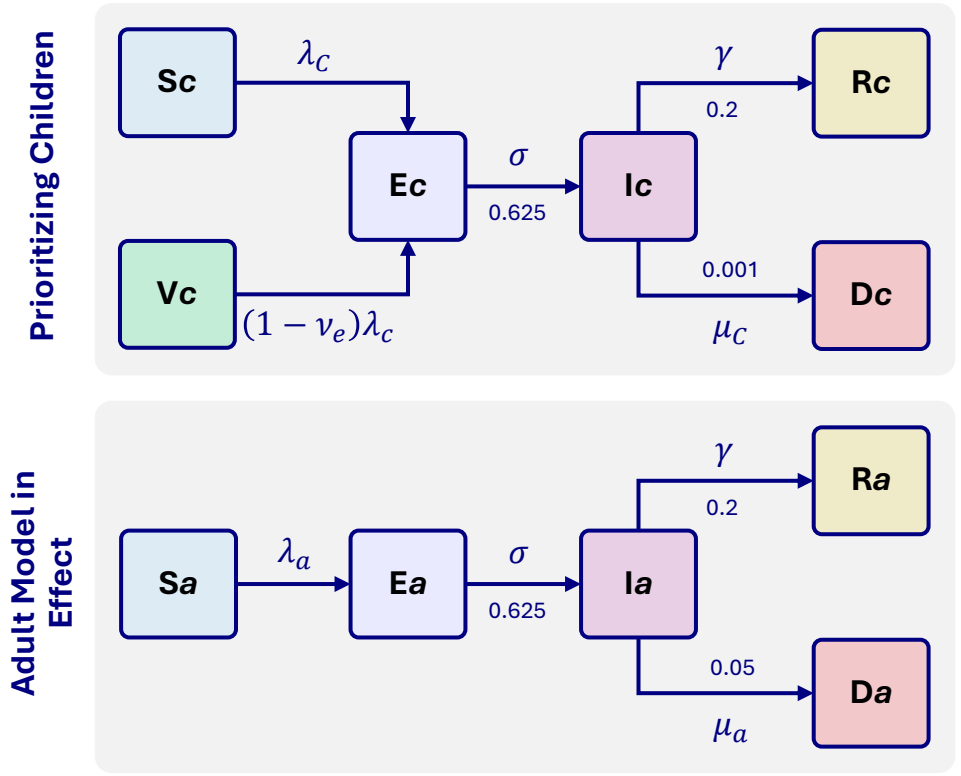
Modelling dynamics of no & full vaccination can serve as a baseline for prioritization scenarios

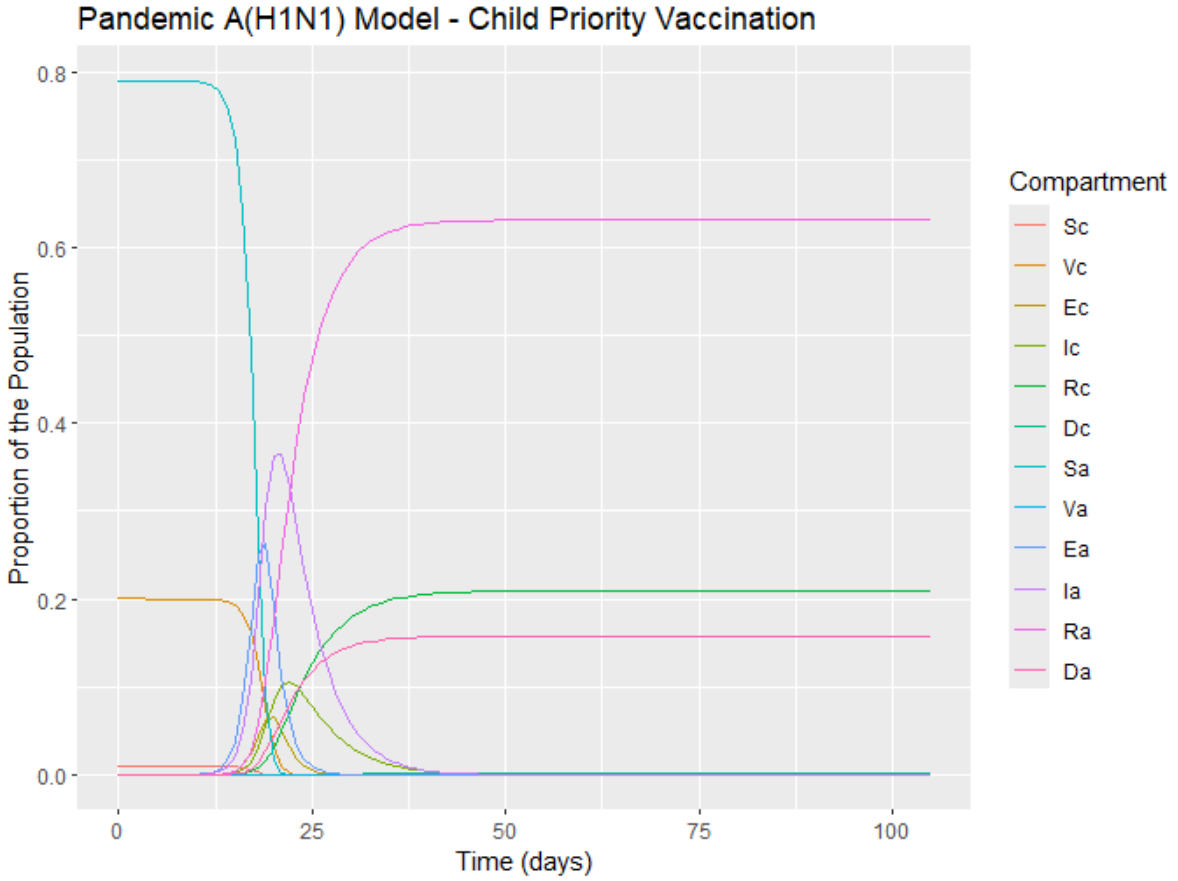




In the first scenario, 20% of the UK population is equivalent to 95% of the child population

$$\lambda_x = \frac{\beta_{xx} I_x}{N_{x_dynamic}} + \frac{\beta_{xy} I_y}{N_{y_dynamic}}$$





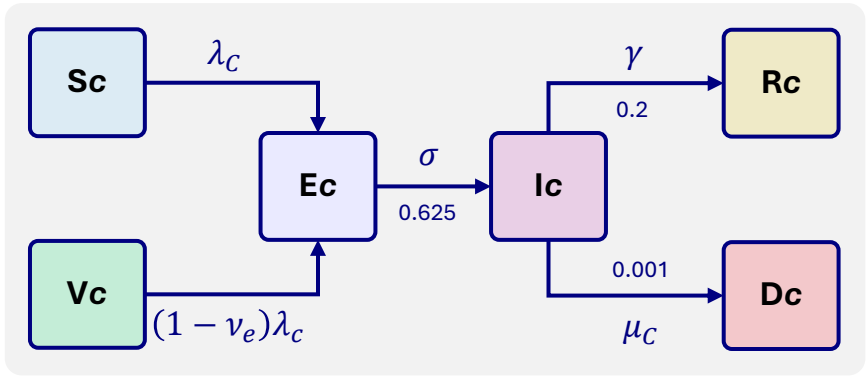
In the first scenario, 20% of the UK population is equivalent to 95% of the child population

$$R_0 = max\left(\frac{1}{\sigma}\left(\frac{\beta_{cc}S_c^{\text{eff}}}{N_c} + \frac{\beta_{ca}S_a^{\text{eff}}}{N_a}\right), \frac{1}{\sigma}\left(\frac{\beta_{aa}S_a^{\text{eff}}}{N_a} + \frac{\beta_{ac}S_c^{\text{eff}}}{N_c}\right)\right)$$

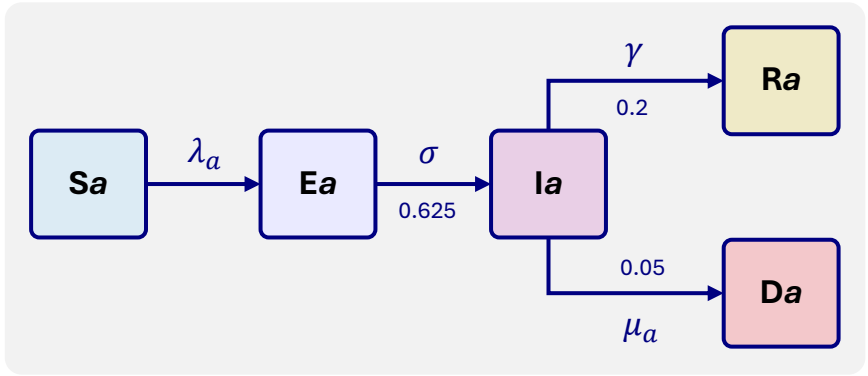
The effective proportion of children that can be infected is:

$S_c^{\text{eff}} = (0.05 + 0.95(1 - 0.6))N_c = (0.43)N_c$

Prioritizing Children

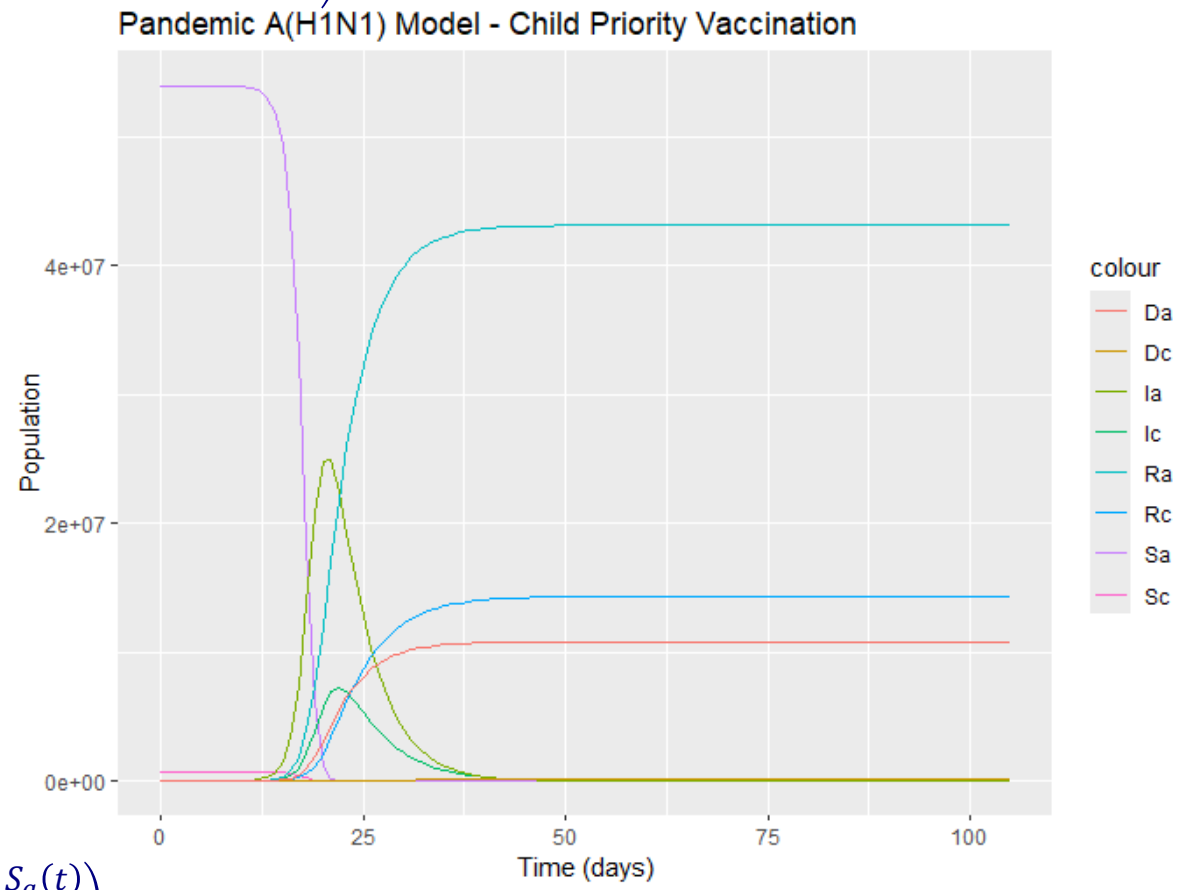


Adult Model in Effect



Since no adults are vaccinated, they remain fully susceptible

$S_a^{\text{eff}} = (1)N_a$



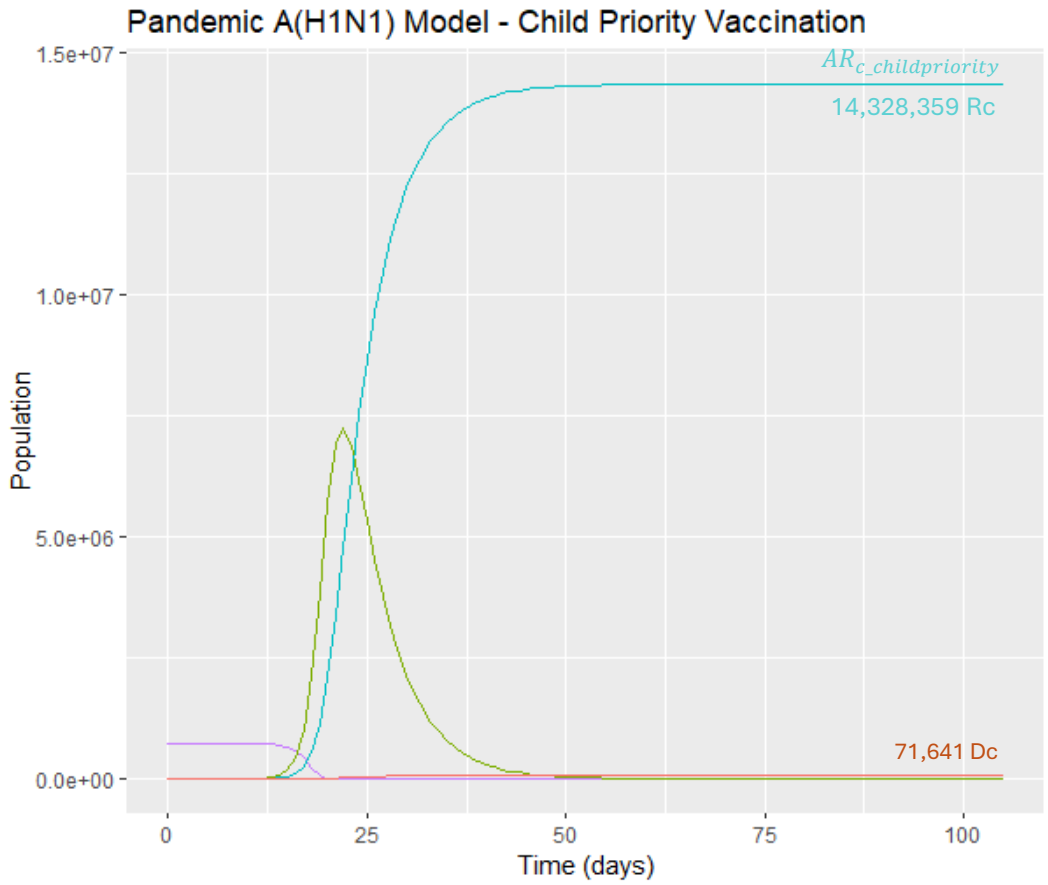
$$R_e = R_0\left(\frac{S_c(t)}{N_C} + \frac{S_a(t)}{N_a}\right)$$

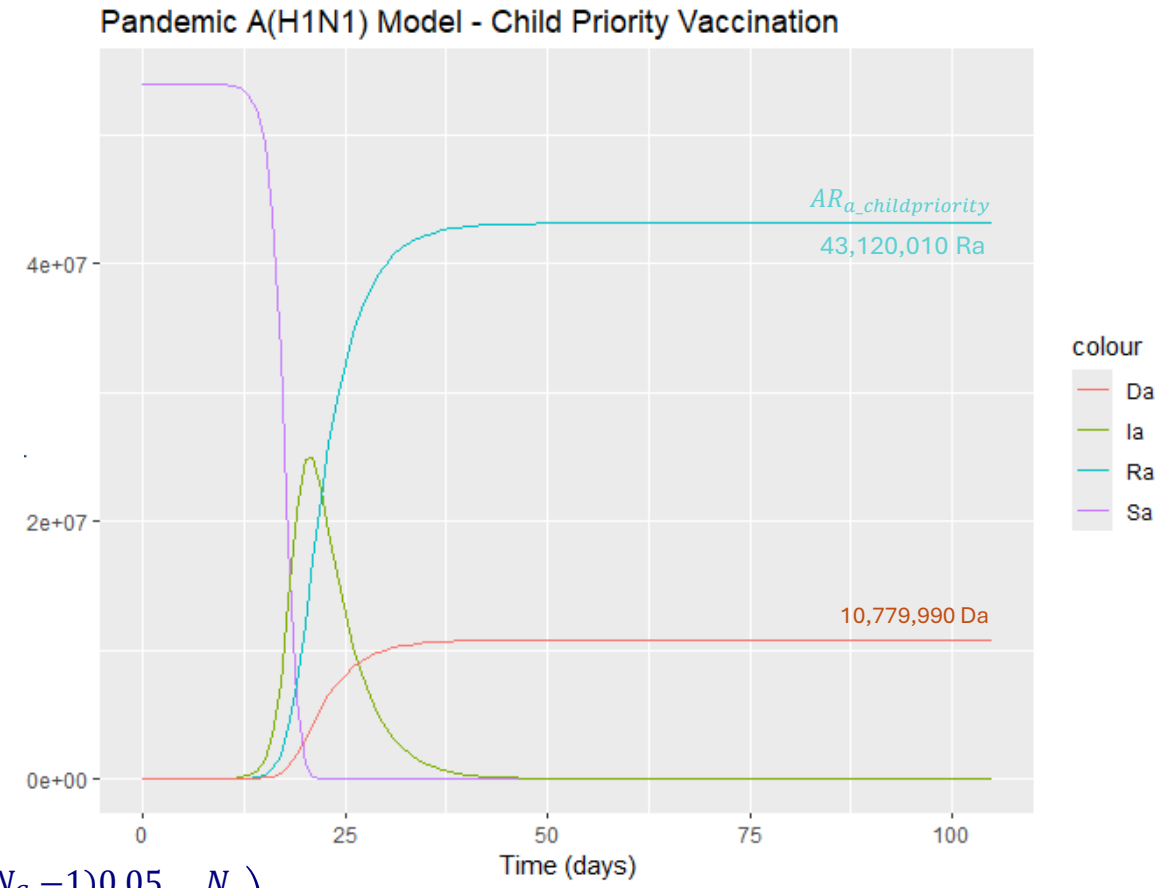
R_e will decline over time as more individuals become infected and recover, reducing the number of susceptible(s)

In the first scenario, 20% of the UK population is equivalent to 95% of the child population

$$R_0 = max(4.435, 4.454)$$

$$R_{0_childpriority} = 4.454$$





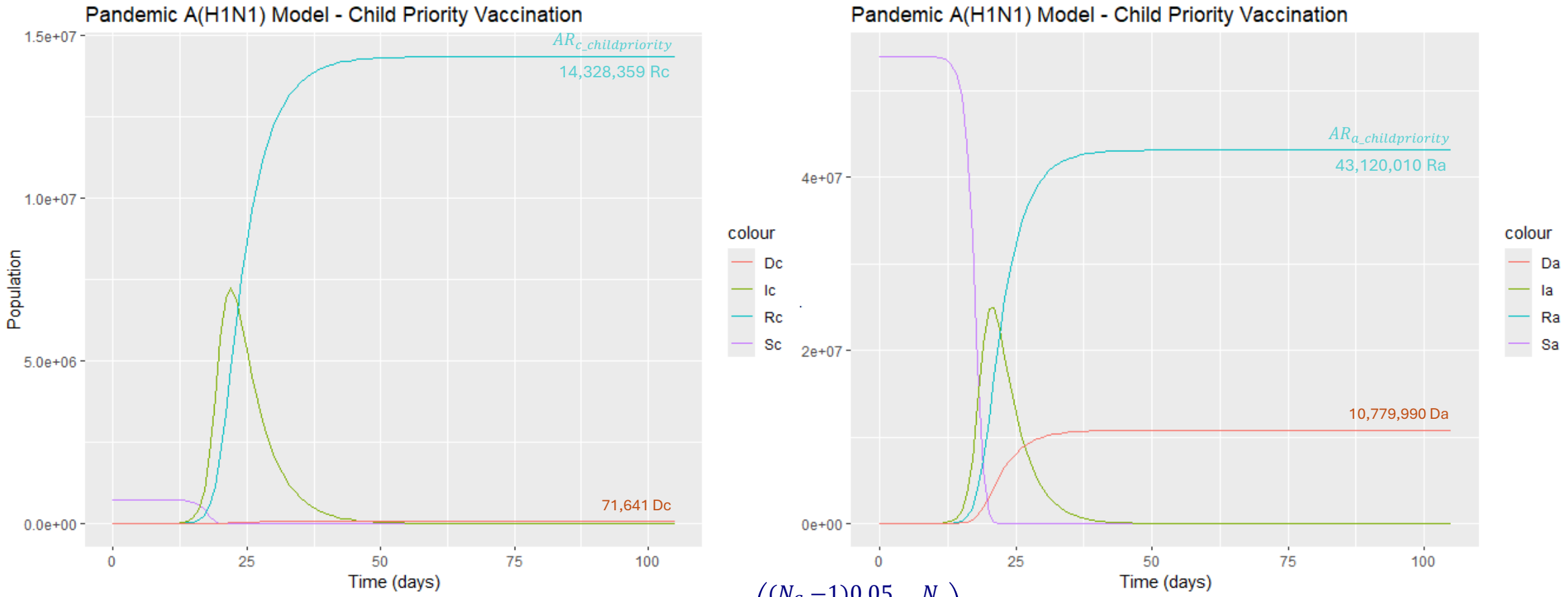
$$At\ t = 0, R_e = 4.454\left(\frac{(N_C - 1)0.05}{N_C} + \frac{N_a}{N_a}\right)$$

$$R_{e(0)_childpriority} = 4.677$$

In the first scenario, 20% of the UK population is equivalent to 95% of the child population

$$Herd\ Immunity\ Threshold\ (HIT) = 1 - \left(\frac{1}{R_0}\right)$$

$$HIT_{childpriority} = 0.775$$

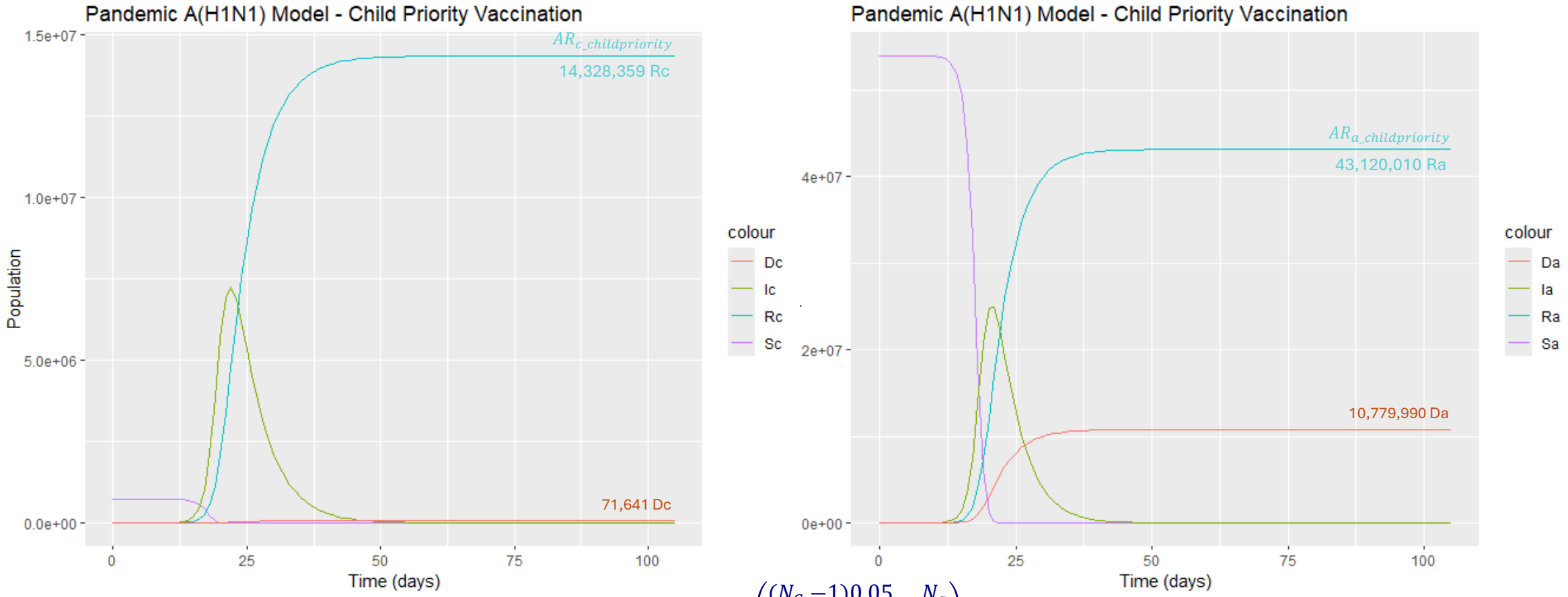


$$At\ t = 0, R_e = 4.454\left(\frac{(N_C - 1)0.05}{N_C} + \frac{N_a}{N_a}\right)$$

$$R_{e(0)_childpriority} = 4.677$$

In the first scenario, 20% of the UK population is equivalent to 95% of the child population

77.5% of the population need to be immune (via infection or vaccination) to prevent sustained transmission

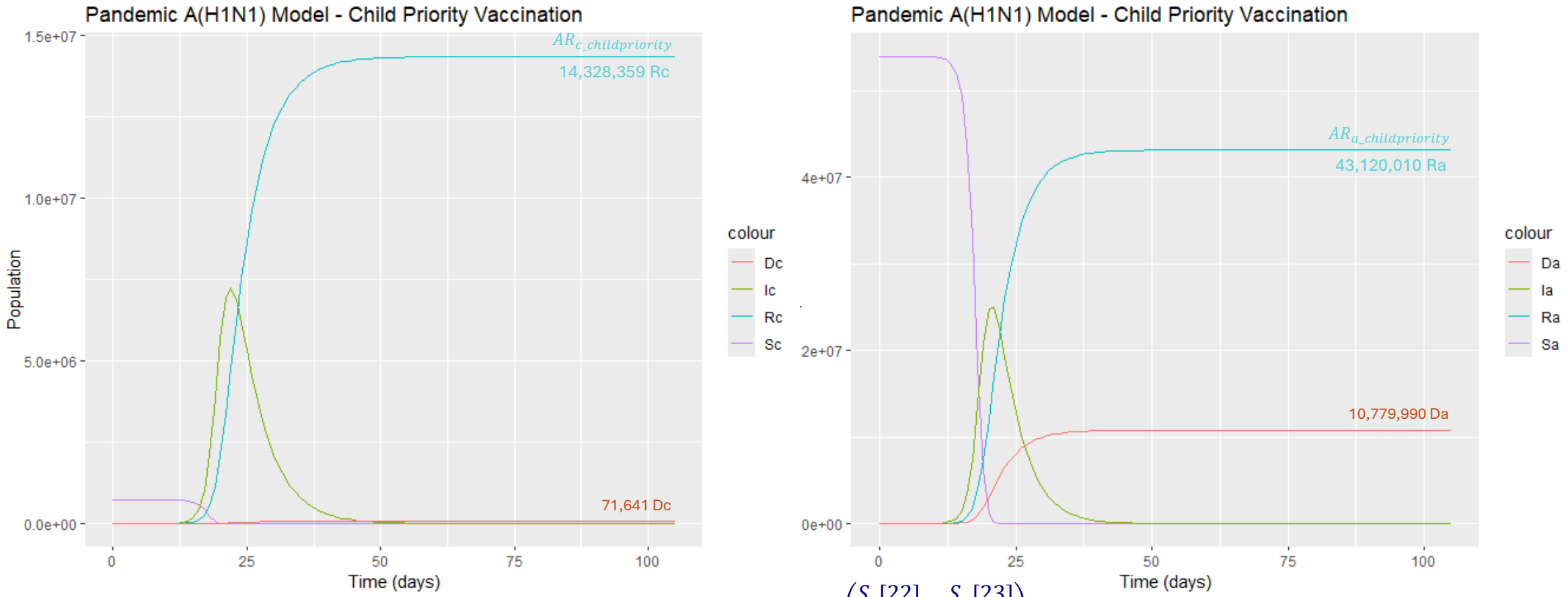


$$At\ t = 0, R_e = 4.454\left(\frac{(N_C - 1)0.05}{N_C} + \frac{N_a}{N_a}\right)$$

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In the first scenario, 20% of the UK population is equivalent to 95% of the child population

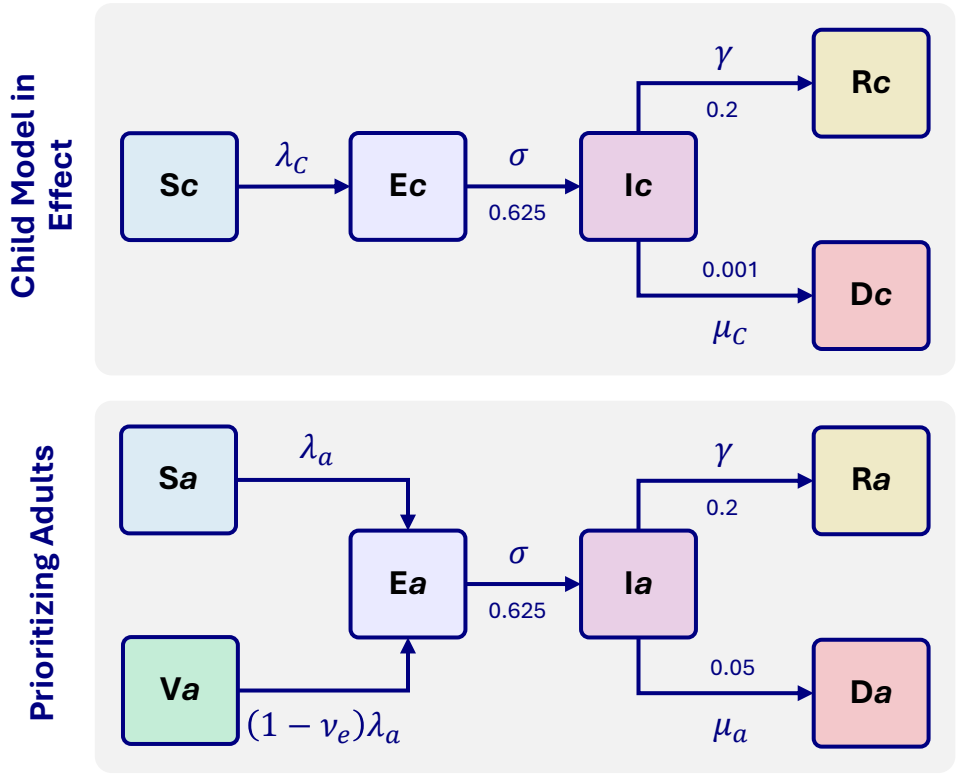
R_e crosses from above 1 to below 1 when $(I_c + I_a)$ reaches its maximum; it is here, at the peak of epidemic, that herd immunity has been reached

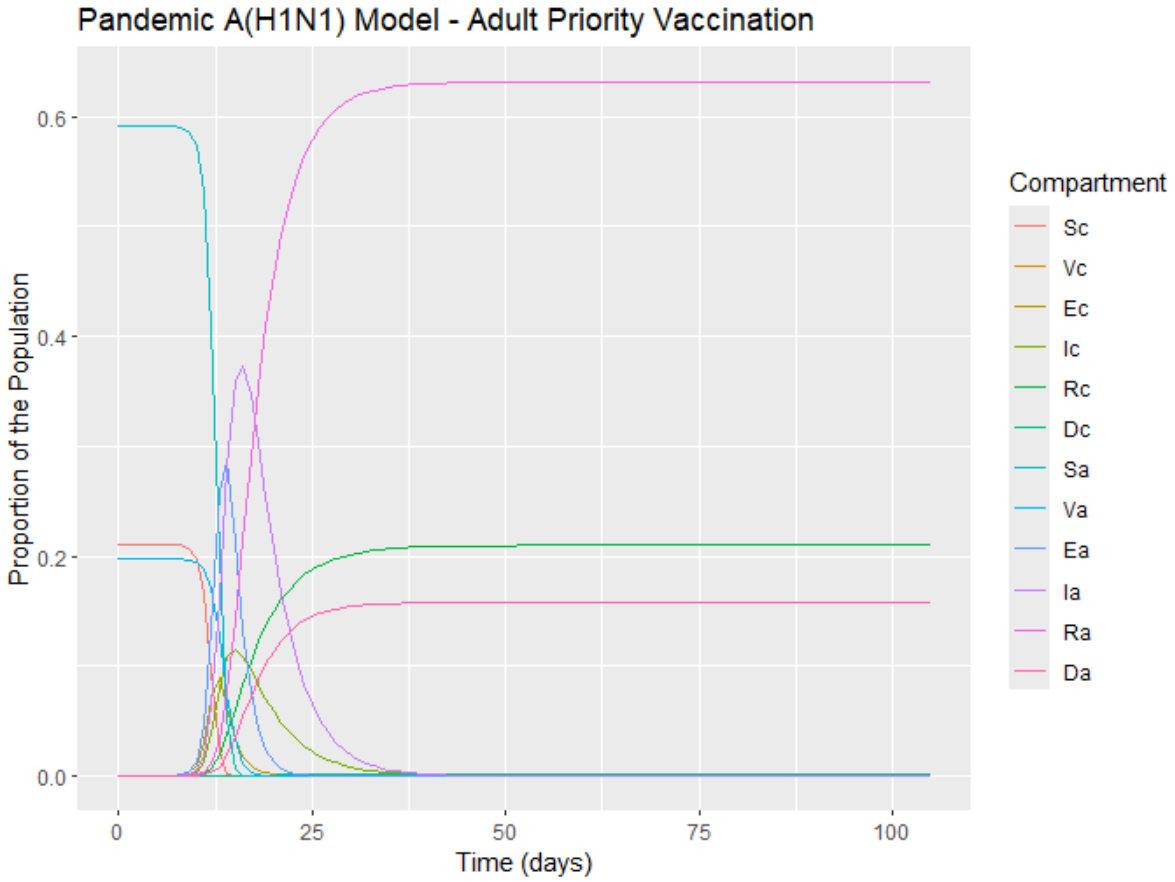


$$At\ [t = epidemic\ peak], R_e = 4.454\left(\frac{S_c[22]}{N_C} + \frac{S_a[23]}{N_a}\right)$$

$$R_{e(peak)_childpriority} = 0.0095$$

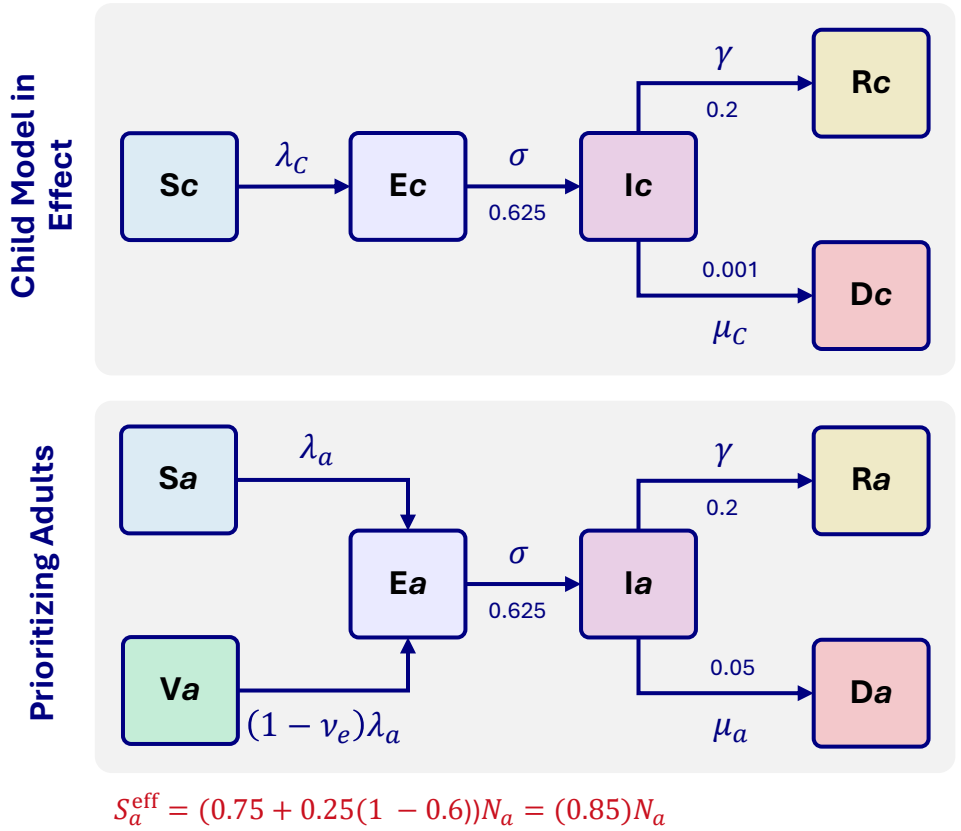
In the second scenario, 20% of the UK population is equivalent to 25%% of the adult population



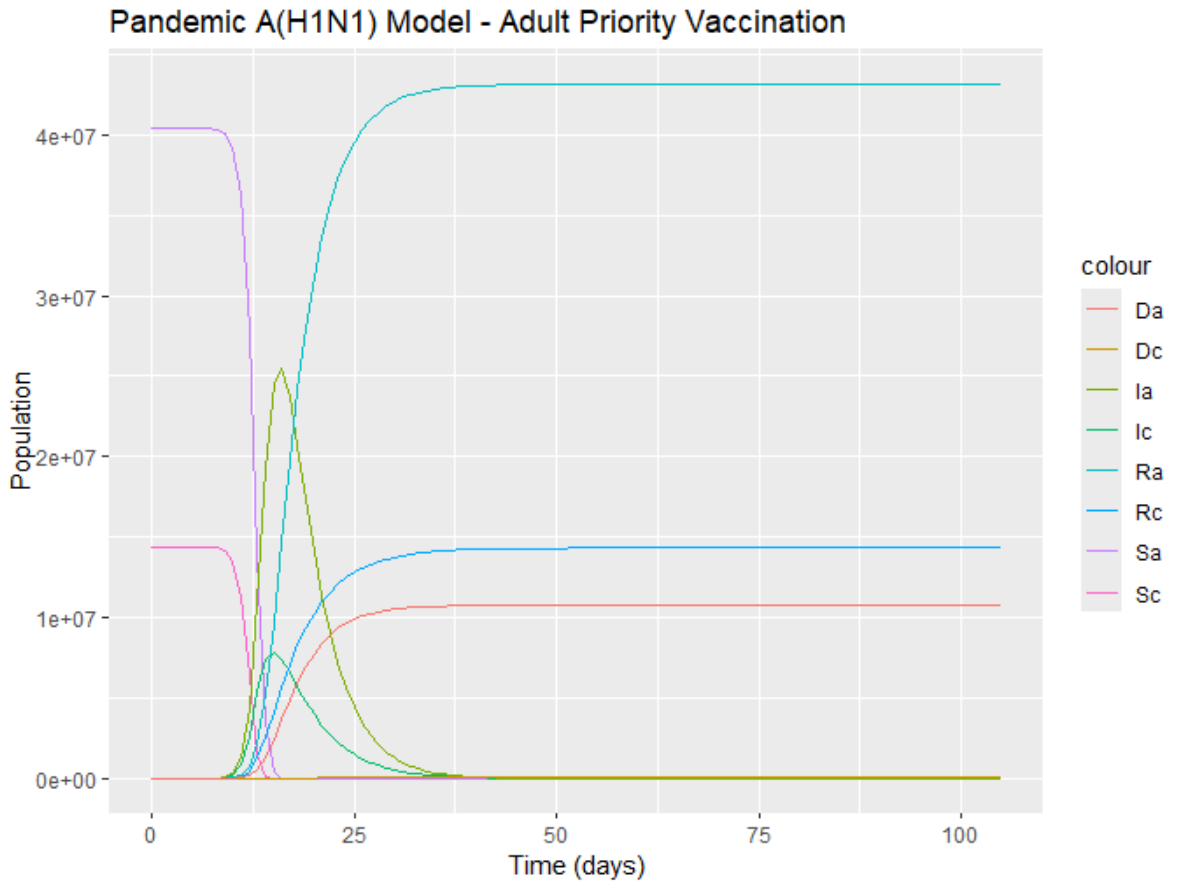


In the second scenario, 20% of the UK population is equivalent to 25%% of the adult population

$S_c^{\text{eff}} = (1)N_c$



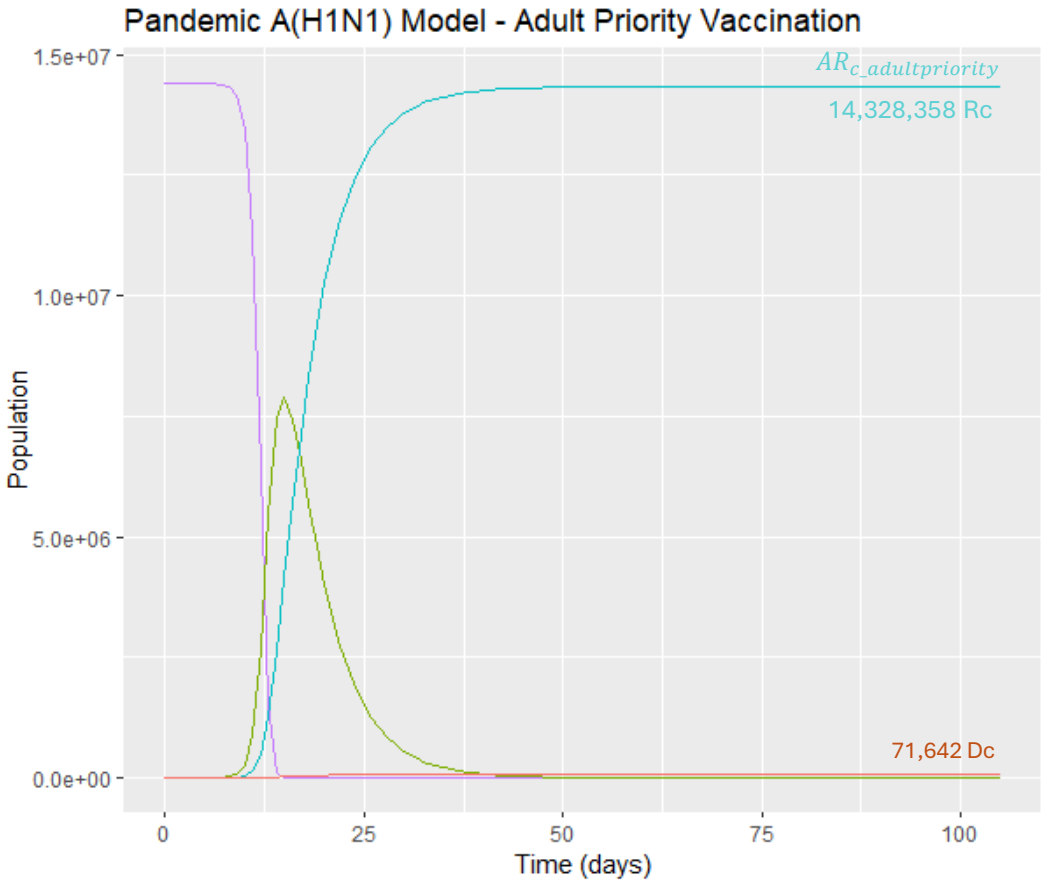
$S_a^{\text{eff}} = (0.75 + 0.25(1\ - 0.6))N_a = (0.85)N_a$

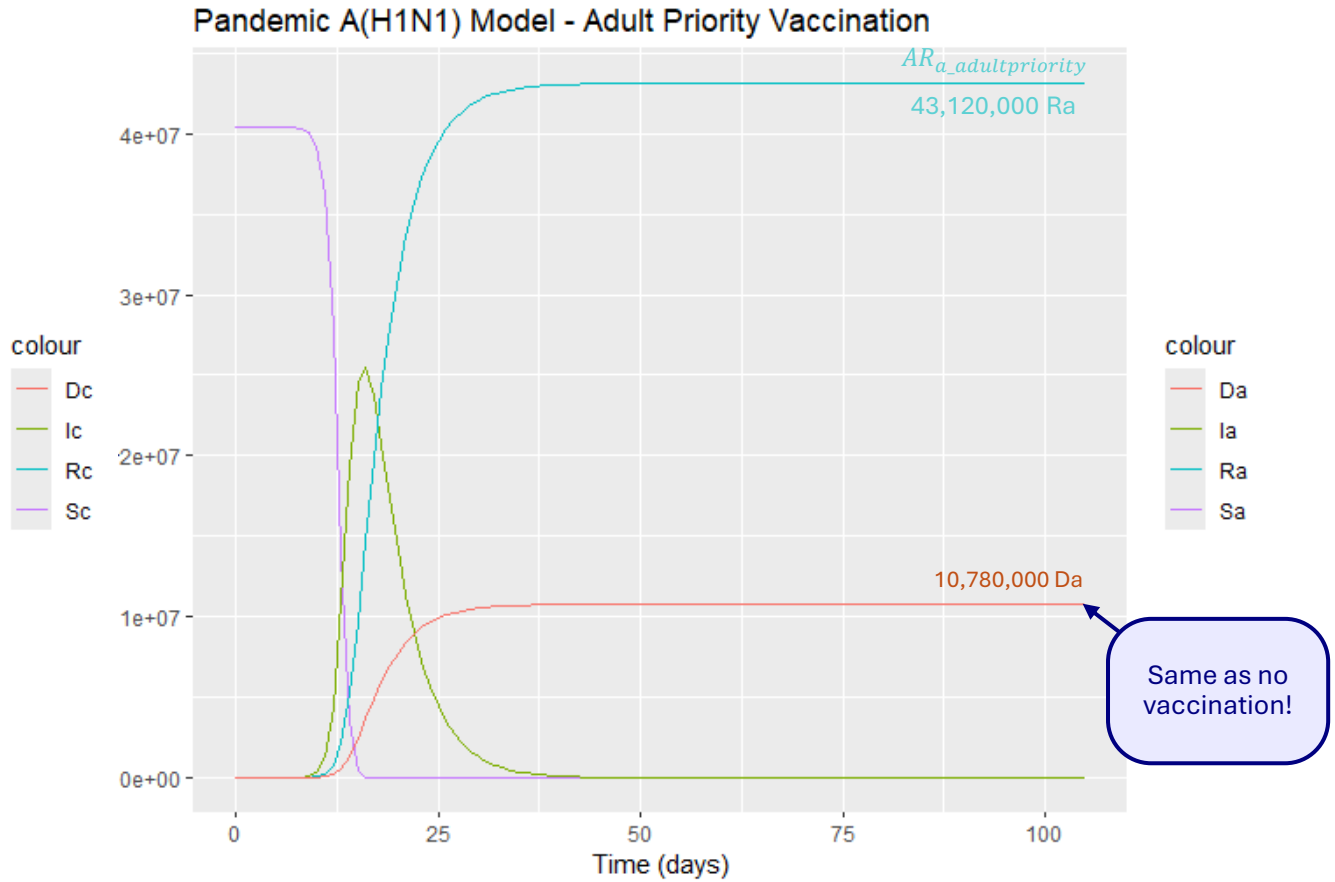


In the second scenario, 20% of the UK population is equivalent to 25%% of the adult population

$$R_{0_adultpriority} = 8.237$$

$$HIT_{adultpriority} = 0.879$$





$$At\ t = 0, R_e = 8.237\left(\frac{(N_c - 1)}{N_c} + \frac{(N_a)\ 0.75}{N_a}\right)$$

$$R_{e(0)_adultpriority} = 14.415$$

$$At\ [t = epidemic\ peak], R_e = 8.237\left(\frac{S_c[15]}{N_C} + \frac{S_a[16]}{N_a}\right)$$

$$R_{e(peak)_adultpriority} = 0.0074$$

Results suggest prioritizing children is the less of two 'evils'

Reproduction Numbers & Herd Immunity

- An R_0 of x means on average, one infected individual causes x new infections in a completely susceptible population

$$R_{0_childpriority} = 4.454$$
$$R_{0_adultpriority} = 8.237$$

- An R_e of y means on average, one infected individual causes y new infections at a specific point in time

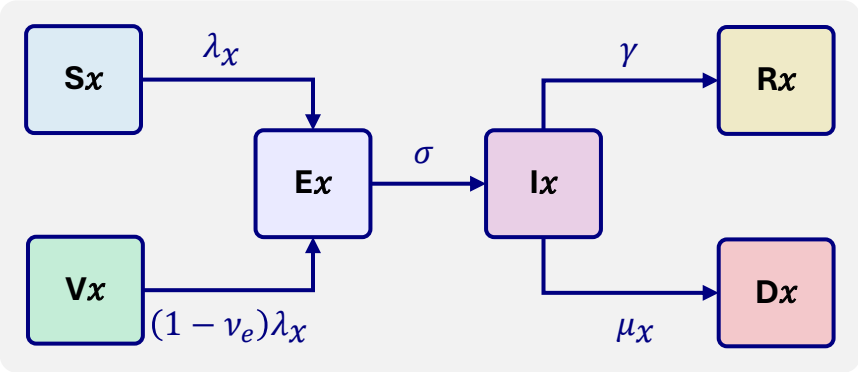
$$R_{e(0)_childpriority} = 4.677$$
$$R_{e(0)_adultpriority} = 14.415$$

- Both values for R_0 and $R_{e(0)}$ are well above 1, meaning the infection spread exponentially, leading to an epidemic as we've seen
- In both scenarios, I_c fills at a faster rate than I_a, reaching peak of epidemic 1 day before; this suggests $FOI_c > FOI_a$ across scenarios
- Epidemic reached its peak in the child-prioritization scenario ~ 1 week after the peak of epidemic in adult-prioritization, suggesting infection spread quicker in the latter scenario

- The R_0 can be used to calculate the herd immunity thresholds, and found that:
- In the child-prioritization scenario, 77.5% of the population need to be immune (via infection or vaccination) to prevent sustained transmission
- In the adult-prioritization scenario, 87.9% will need to be immune

- We know that each child an average contacts 13 people in the population per day, and each adult contacts 7
- Although each child is responsible for more daily contacts than each adult, there are far more adults than children in the pool
- At 3.74x the size of the child population, adults are responsible for 377.3 million contacts per day, whereas children are only responsible for 187.2 million
- If we prioritize children (vaccination 95% of them, and considering the vaccine's effectiveness), there are still too many adults that can still spread infection
- If we prioritize adults (vaccinating 25%), we haven't covered enough of them; nor children, who have a higher chance of transmitting disease per individual
- In both scenarios, there isn't enough vaccine available to achieve herd immunity in the UK population
- However, results suggest prioritizing children means less of the population will need to acquire immunity to reach herd immunity

Case Fatality Rates for children & adults are constant across scenarios



$$Survival\ Rate\ (SR)_x = \frac{\gamma}{\mu_x + \gamma}$$

$SR_c = 0.995$

$SR_a = 0.800$

$$Case\ Fatality\ Rate\ (CFR)_x = \frac{\mu_x}{\mu_x + \gamma}$$

$CFR_c = 0.005$

$CFR_a = 0.200$

$$Average\ Duration\ of\ Infectious\ Period_x = \frac{1}{\mu_x + \gamma}$$

$Average\ Duration\ of\ Infectious\ Period_c = 4.975$ days

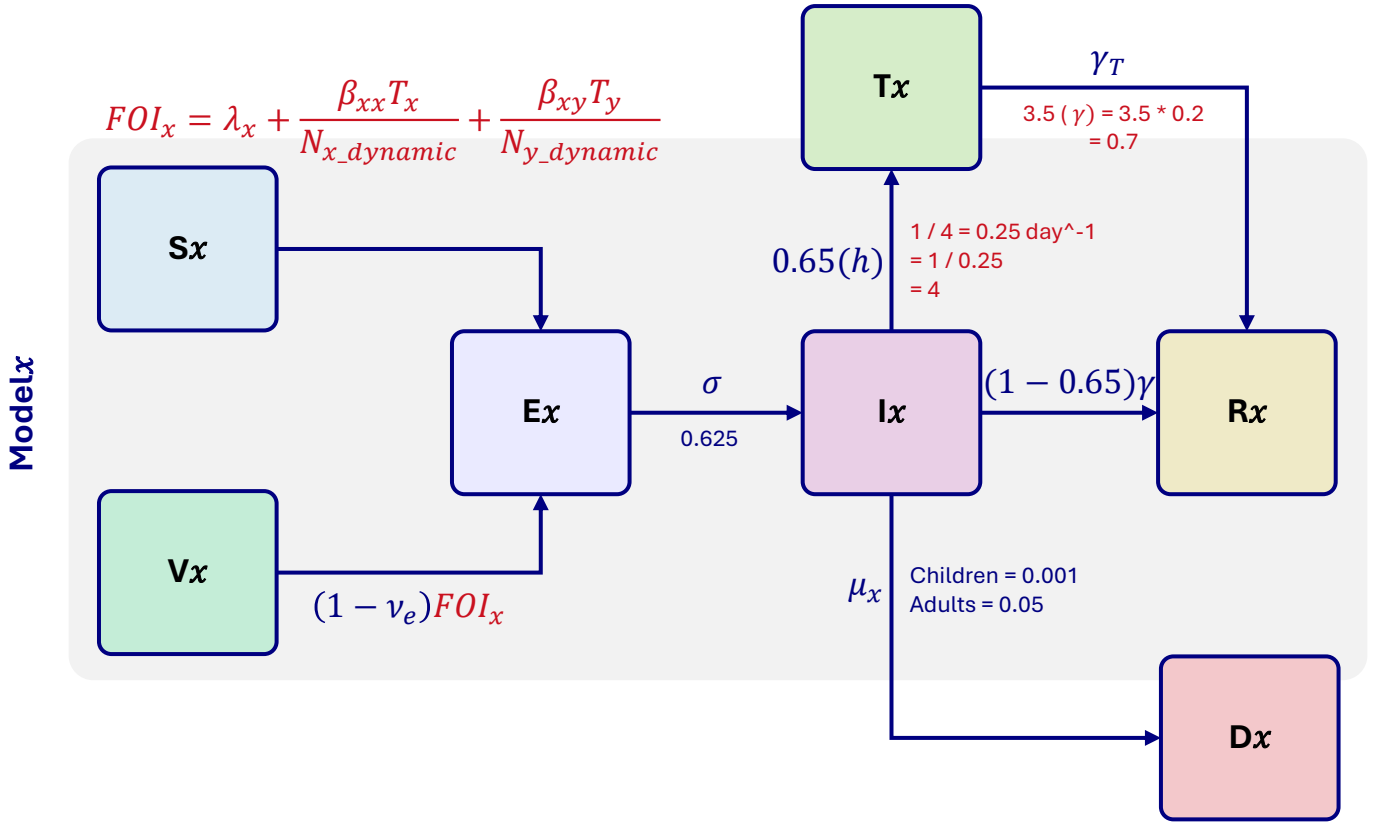
$Average\ Duration\ of\ Infectious\ Period_a = 4.000$ days

Case Fatality Rates for children & adults are constant across scenarios

What's Up with the Death Counts?

- Across scenarios, the hazard of recovering is higher than the hazard of death, because $\gamma > \mu_x$
- In both scenarios, the final death count for adults was higher than that of children, because $CFR_a > CFR_c$
- Final death counts were slightly lower in adult prioritization scenario than no vaccination
- Final death counts were slightly lower in child prioritization scenario than adult prioritization scenario
- Collectively, the results suggest the vaccine doesn't act to reduce the chance of recovery once infected in all vaccinees directly; it acts by reducing the proportion of individuals that will become infected
- It would be interesting to see transmission dynamics if we were to increase the vaccine's coverage, or introduce treatment which acts directly to impact the recovery rate

We can introduce treatment using the same model to visualize changes in dynamics

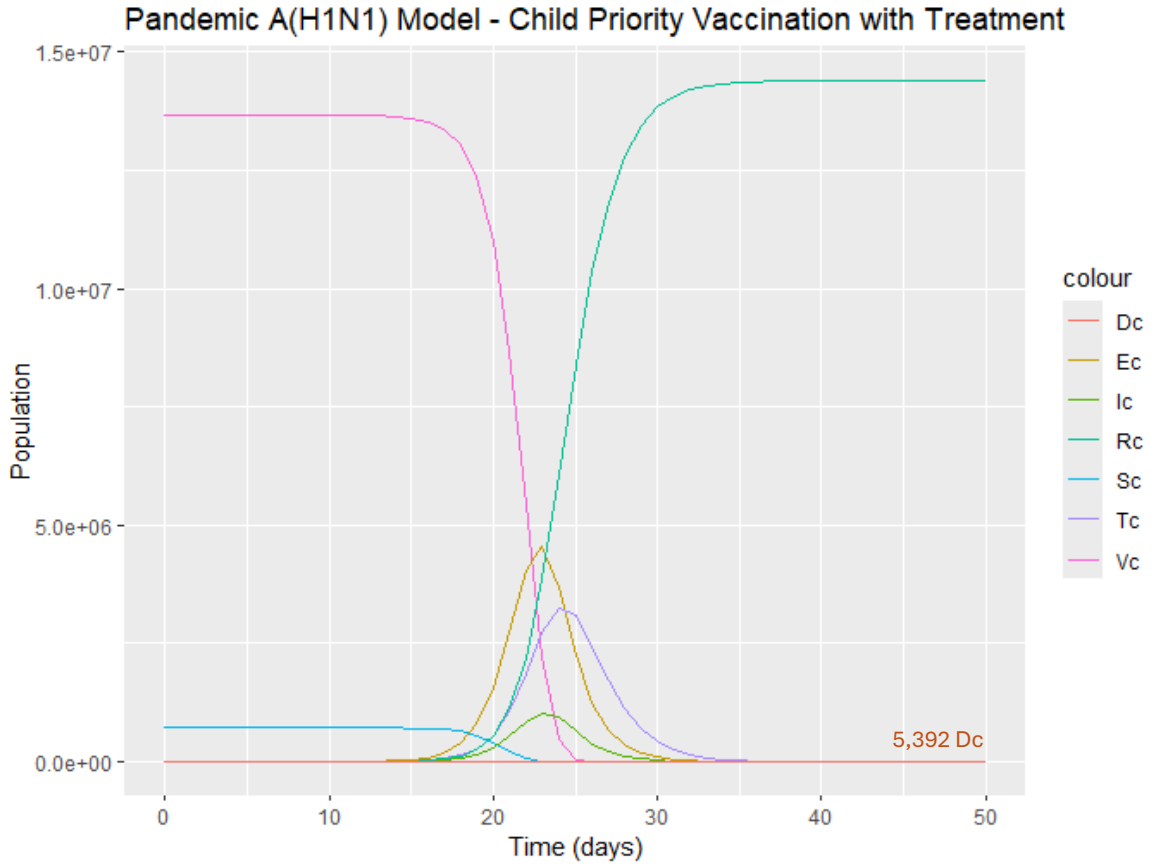


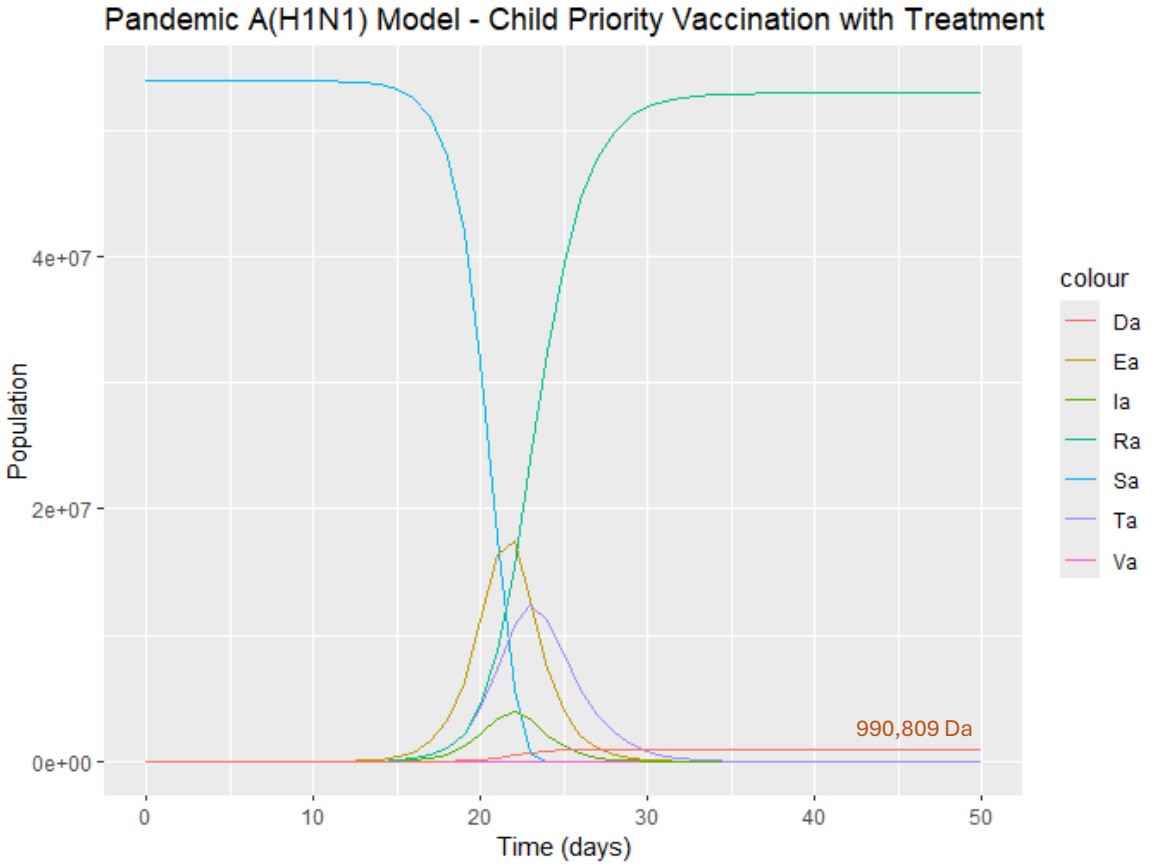
The New Model: Vaccination + Antivirals

- We will model both the child-priority and adult-priority scenarios, but with the introduction of antivirals
- We will assume the antiviral will be taken by 65% of the infected population 4 days post-symptom onset
- We will assume partial effectiveness at blocking transmission, and that treated people recover 3.5x faster than those untreated

We can introduce treatment using the same model to visualize changes in dynamics

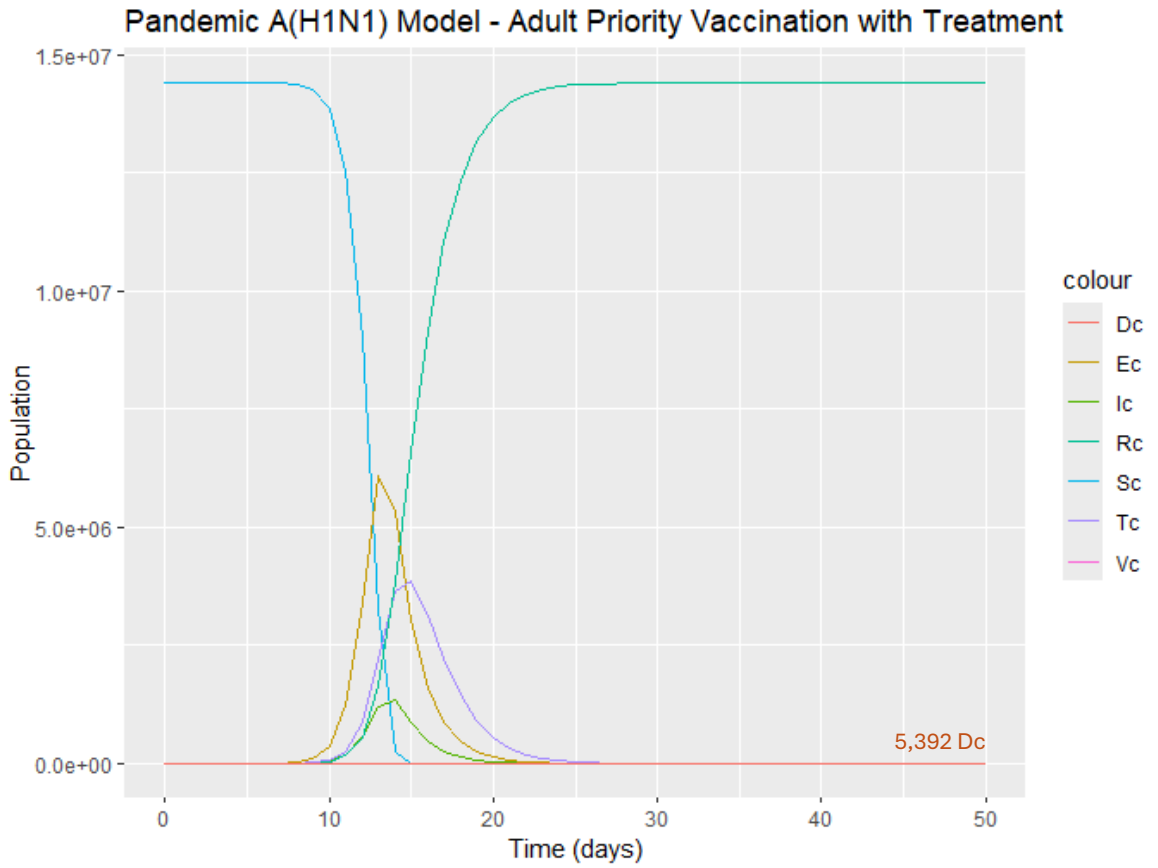
We can see quite the reduction in death with the introduction of antivirals!

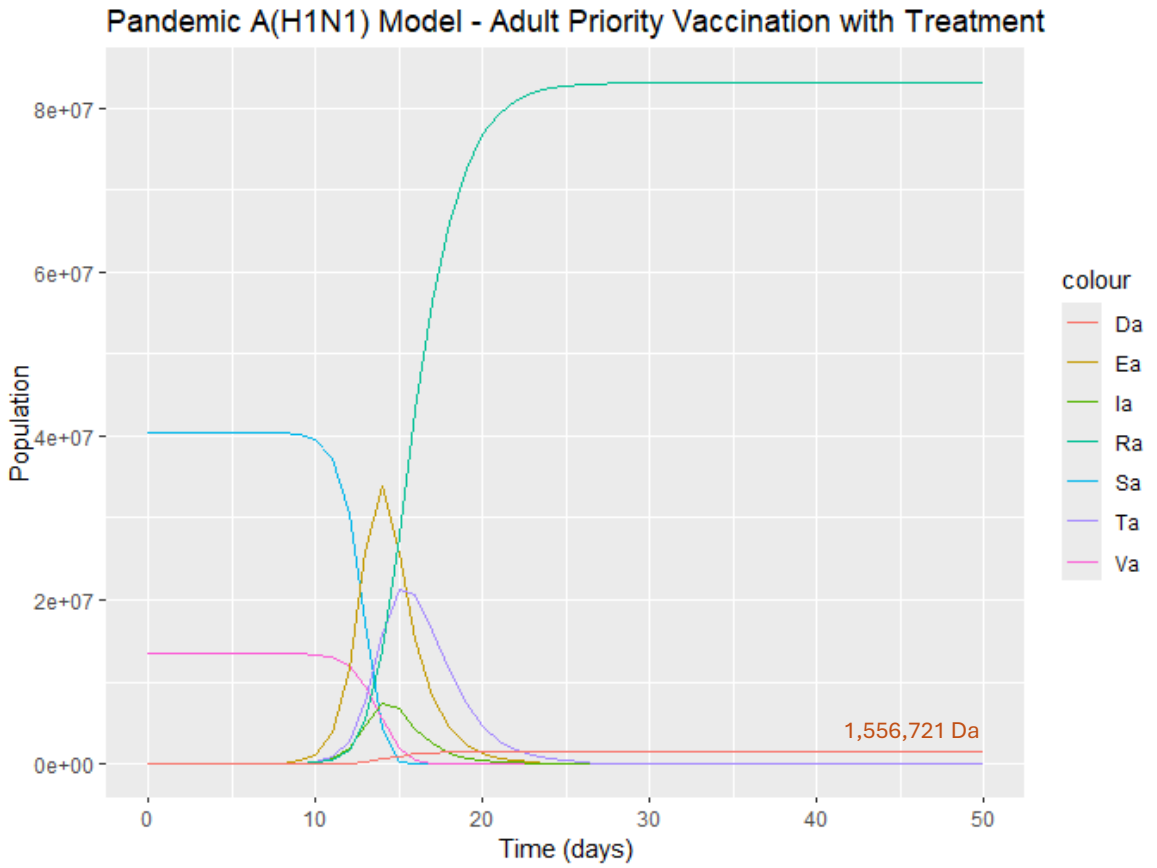




This represents a ~91% reduction in death compared to our best-case scenario (child-prioritization)

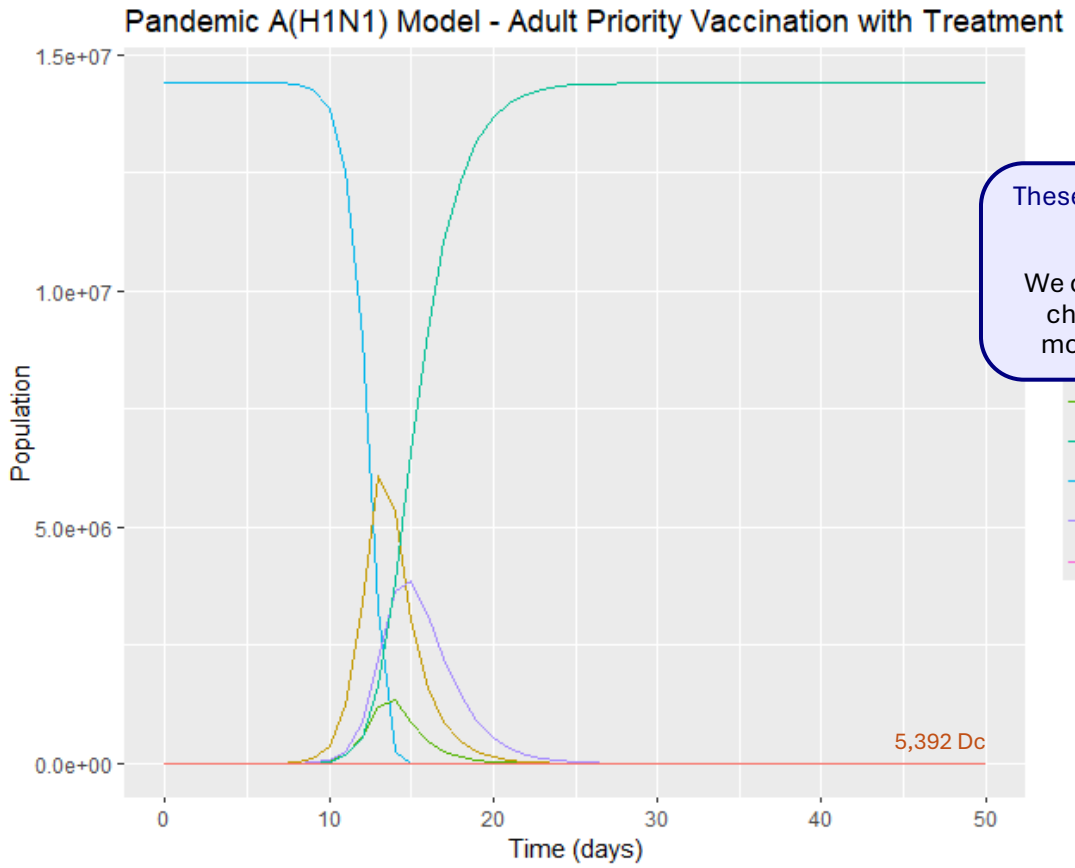
We can introduce treatment using the same model to visualize changes in dynamics

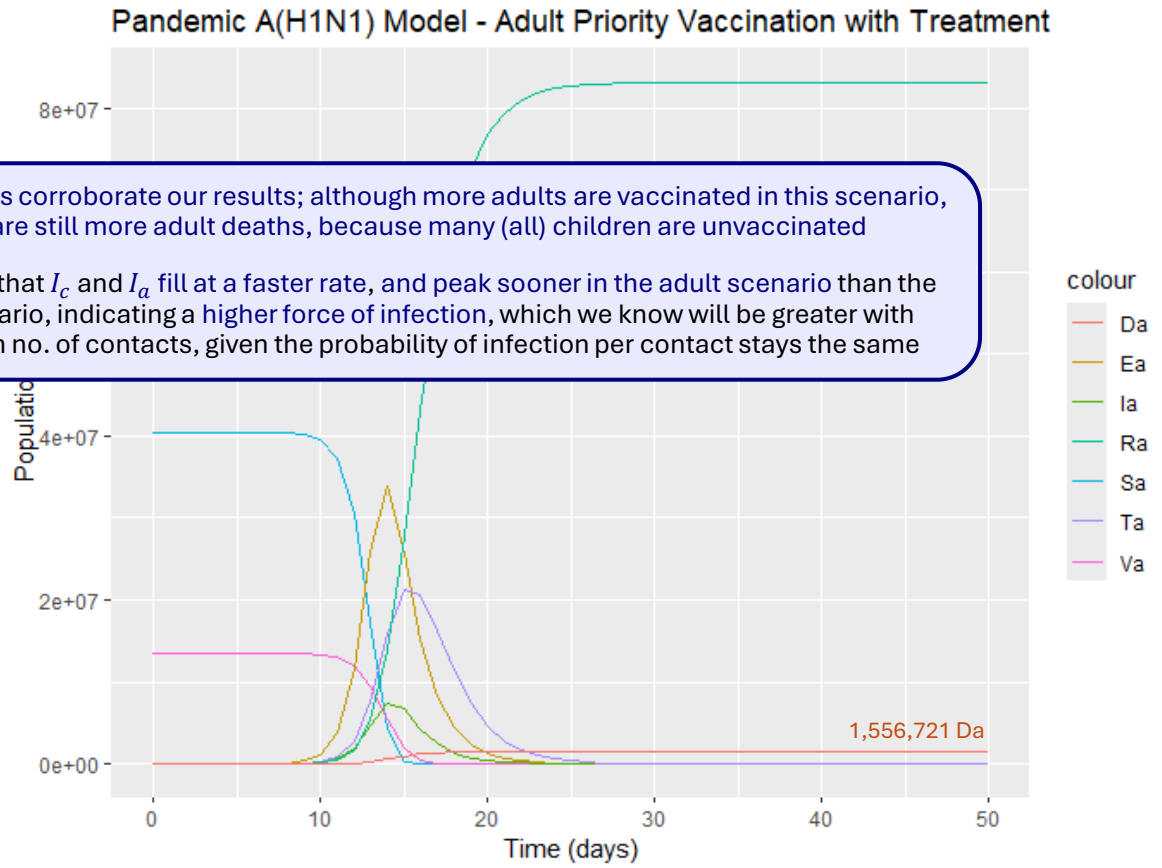




This represents an ~85.6% reduction in death compared to our best-case scenario (child-prioritization)

We can introduce treatment using the same model to visualize changes in dynamics





These findings corroborate our results; although more adults are vaccinated in this scenario, there are still more adult deaths, because many (all) children are unvaccinated

We can see that I_c and I_a fill at a faster rate, and peak sooner in the adult scenario than the child scenario, indicating a higher force of infection, which we know will be greater with more mean no. of contacts, given the probability of infection per contact stays the same

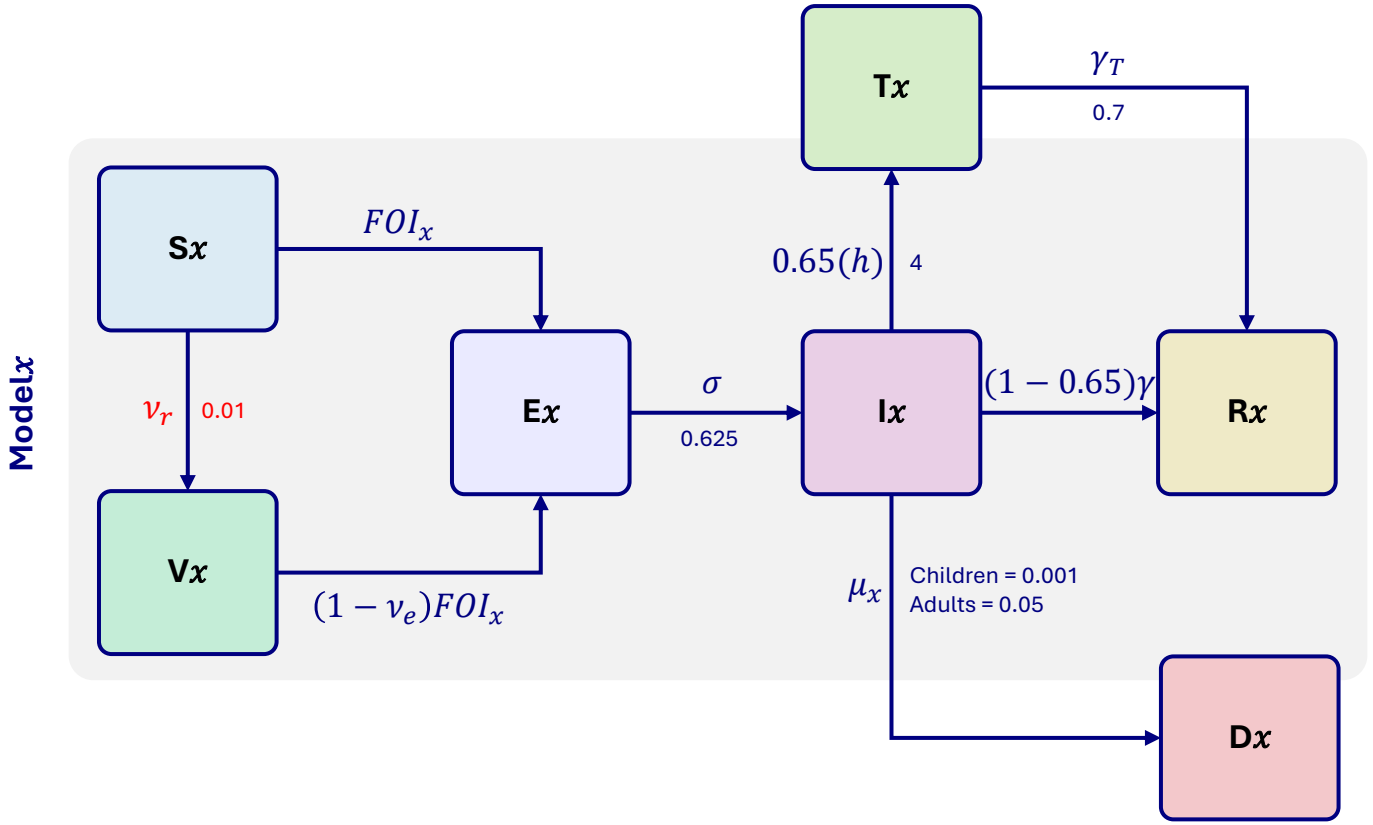
This represents an ~85.6% reduction in death compared to our best-case scenario (child-prioritization)

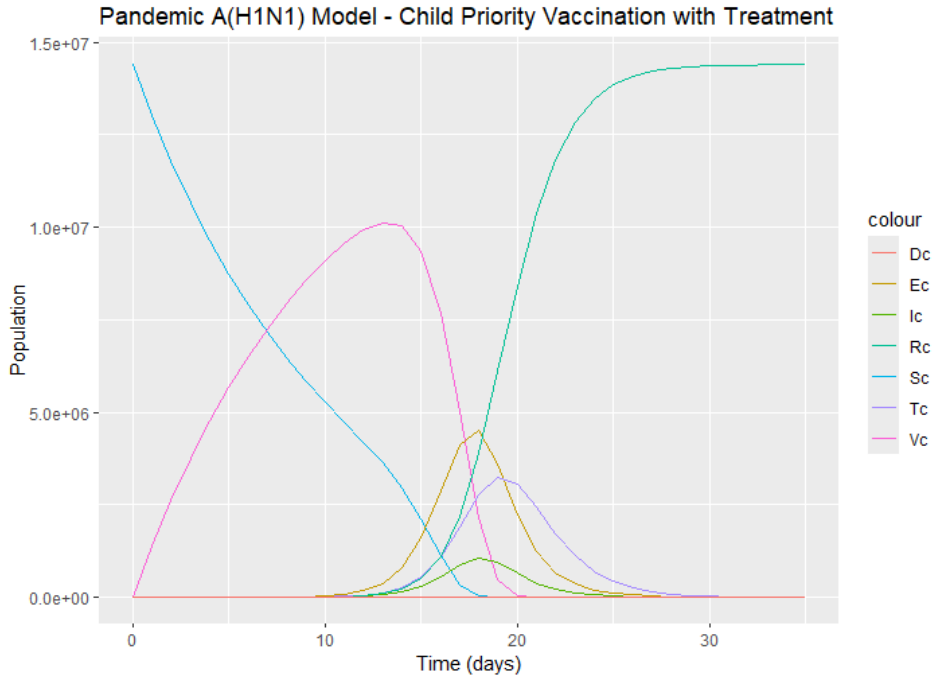
There are a few things we could do to strengthen our analysis

What can we do to elucidate a clearer answer to out research question?

- We could look at actual HRGs (children under 5 (esp. < 2), and adults 65 or older)
- We could reassess scenarios where both groups have an equal number of people in their populations
- We could model best and worst-case scenarios for each prioritization scenario, factoring in SDs on rates and parameters (e.g. latency periods, etc)
- We could implement different recovery rates, given the literature suggests adults tend to recover more quickly from influenza than children
- We could incorporate a vaccination rate into our model, given population vaccination happens over time

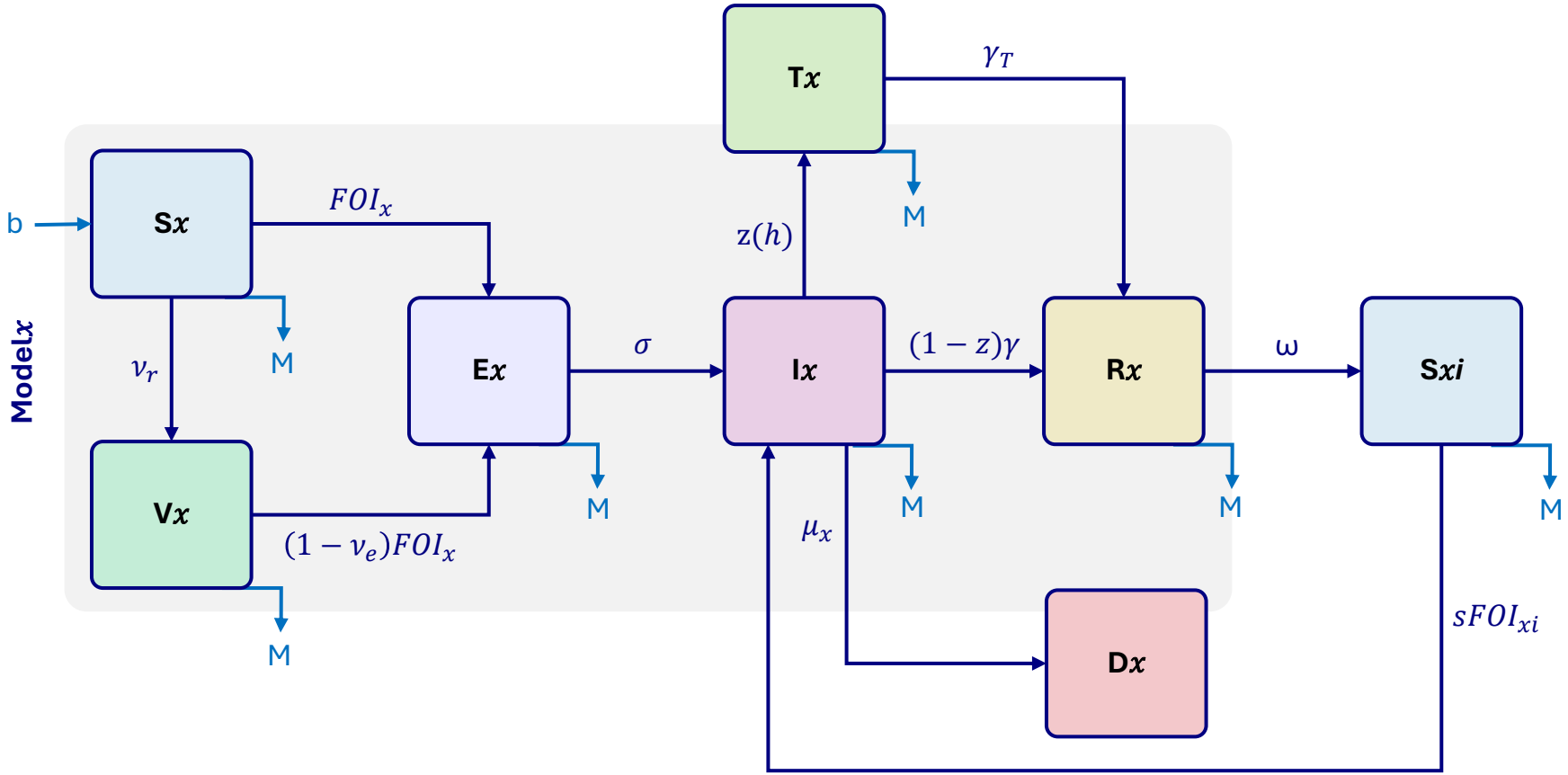
We can account for the rate of vaccination for a more realistic approach





In which case we'd expect it to look like this when prioritizing children, with the full child population starting off in Sc

We can account for the rate of vaccination for a more realistic approach



If we wanted to be ambitious…

- There is waning immunity; individuals retain some protection and can become partially susceptible within the outbreak ($s < 1$)
- The population is not closed; there are births and deaths due to natural causes

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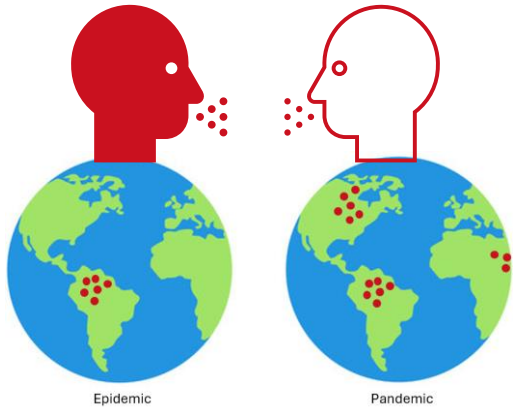
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Thank You

Any Questions?

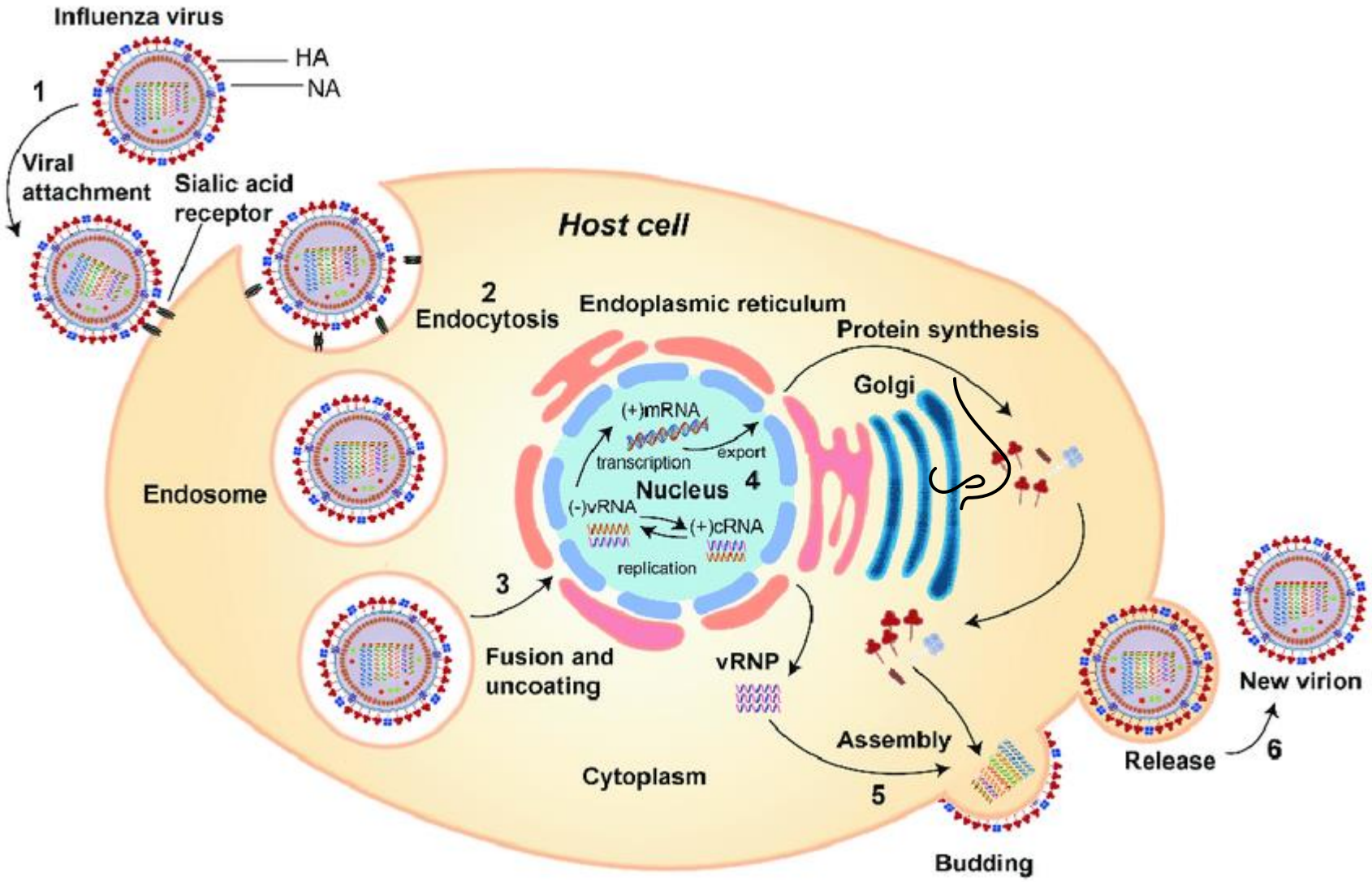
Infectious Disease Modelling Team
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We appreciate your continued attention.



Appendix A

Type-A Influenzas' lifecycle and its symptoms



Symptoms

- Fever
- Cough
- Sore throat
- Runny or stuffy nose
- Muscle / body aches
- Headaches
- Fatigue
- Gastrointestinal complications